UCI Samueli School of Engineering

Collaboration in Multi-Agent Systems: From Assignment to Learning

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PhD Qualifying Examination

Thursday, May 21st, 2026

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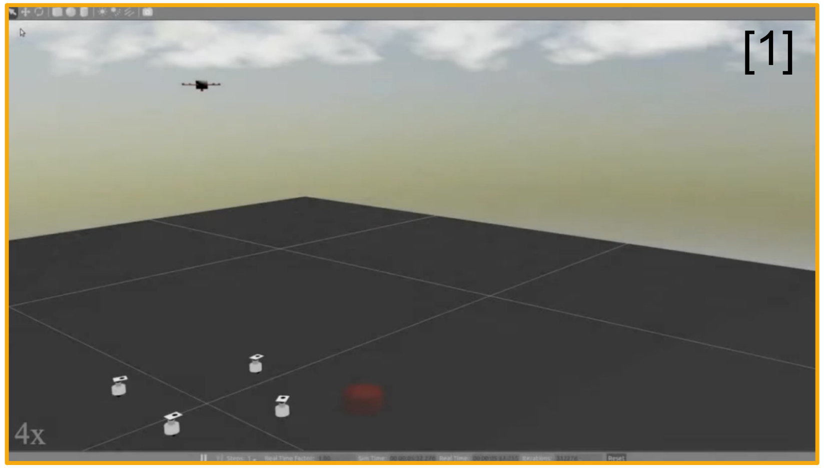
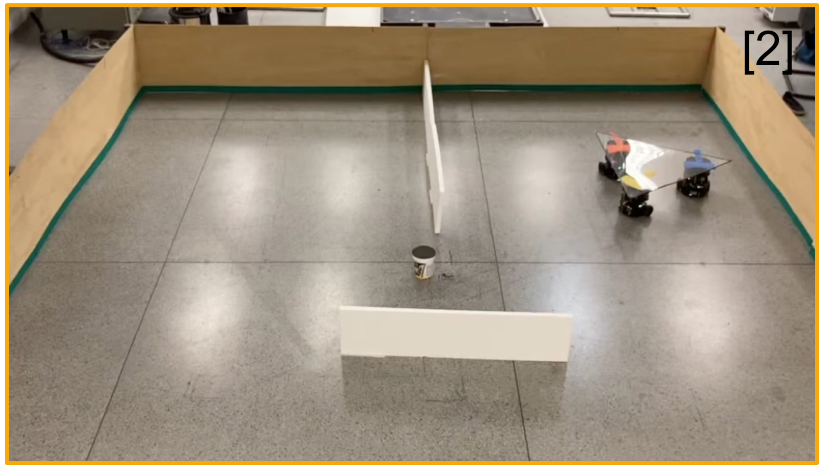
Robot Ecology Lab
UCI Samueli School of Engineering

Motivation

Collaboration | **Assignment** | **Learning**





[1]: Rosa, L., Cognetti, M., Nicastro, A., Alvarez, P., & Oriolo, G. (2015). “Multi-task cooperative control in a heterogeneous ground-air robot team.” IFAC-PapersOnLine, 48(5), 53-58.
[2]: De Sousa, S. K. A., Freire, R. C. S., Carvalho, E. Á. N., Molina, L., Santos, P. C., & Freire, E. O. (2022). Two-layers workspace: A new approach to cooperative object transportation with obstacle avoidance for multi-robot system. IEEE Access, 10, 6929-6939.
[3]: Fu, B., Smith, W., Rizzo, D., Castanier, M., Ghaffari, M., & Barton, K. (2022). Learning task requirements and agent capabilities for multi-agent task allocation. arXiv preprint arXiv:2211.03286.

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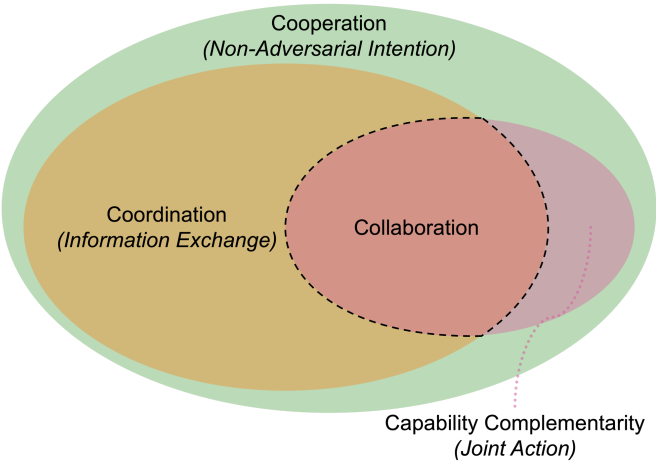
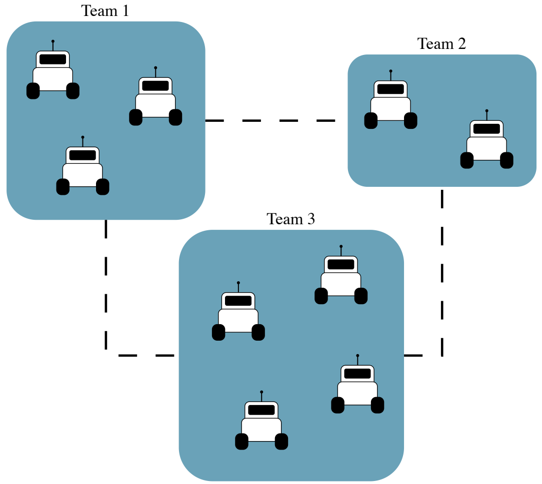
Outline

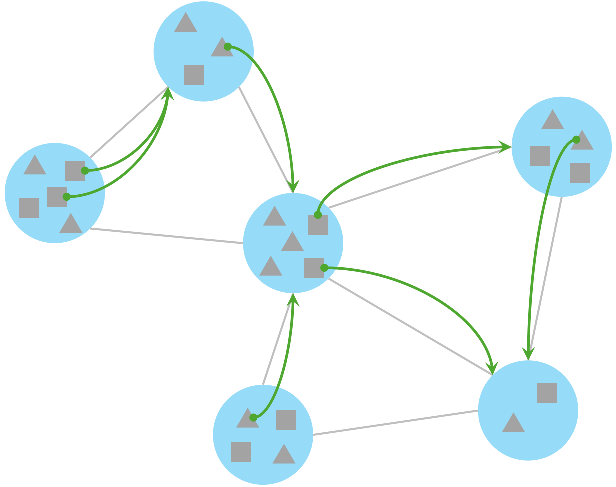
Collaboration in Multi-Agent Systems [1]



Resource Allocation in Multi-Team Systems [2]

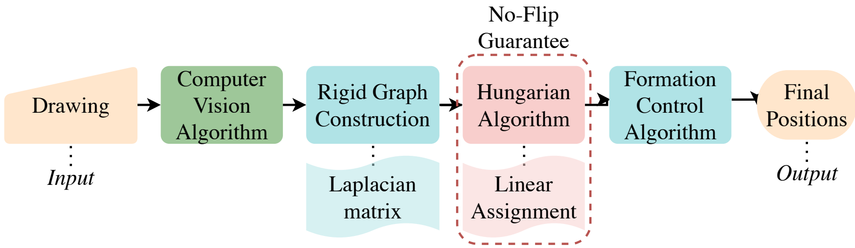


Learning Heterogeneous Altruistic Collaboration [3]



Learning Collaborative Robot Mobility with Model Predictive Scheduling

Human-Swarm Collaboration for Free Multi-Robot Formations



[1]: **Karam, R.**, Nguyen, A. A., Lin, R., Martin, D. R., Morales, D., Butler, B. A., & Egerstedt, M., “Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration.” (submitted to CSM, 2026)

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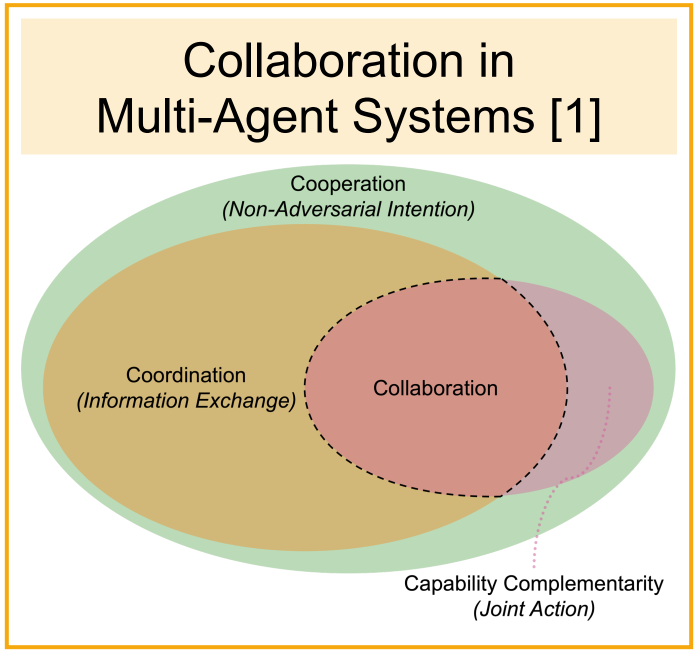

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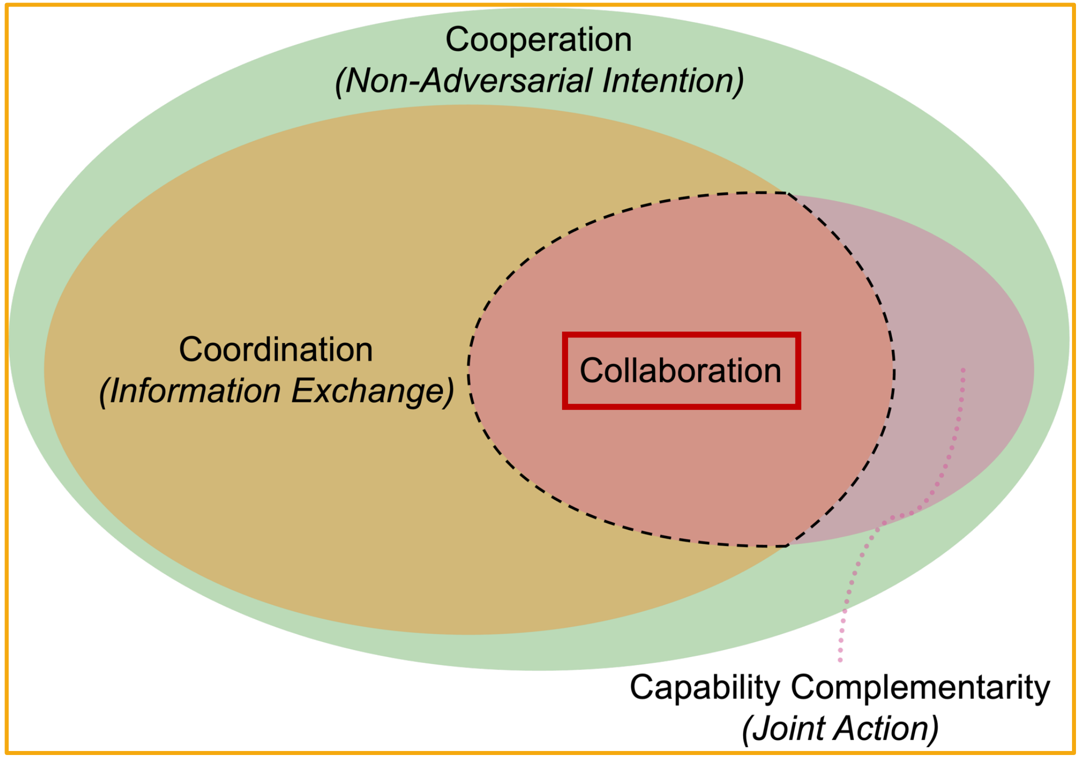
Collaboration in Multi-Agent Systems

Observation: Inconsistency and/or interchangeability of the terminology

- What is *collaboration* in multi-agent systems?
- How is it different than *cooperation* and *coordination*?



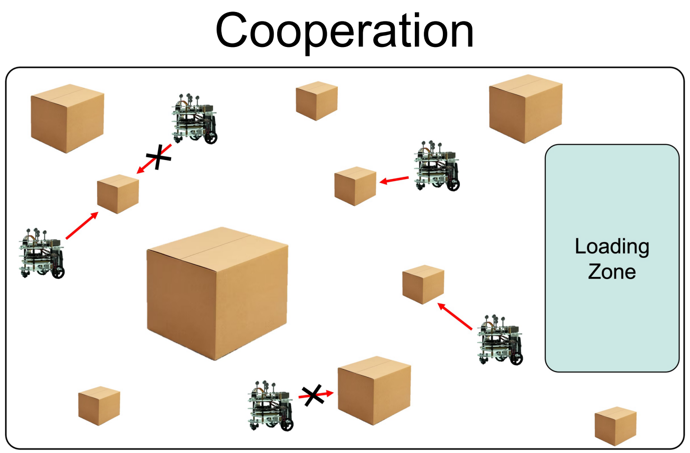
Collaboration in Multi-Agent Systems

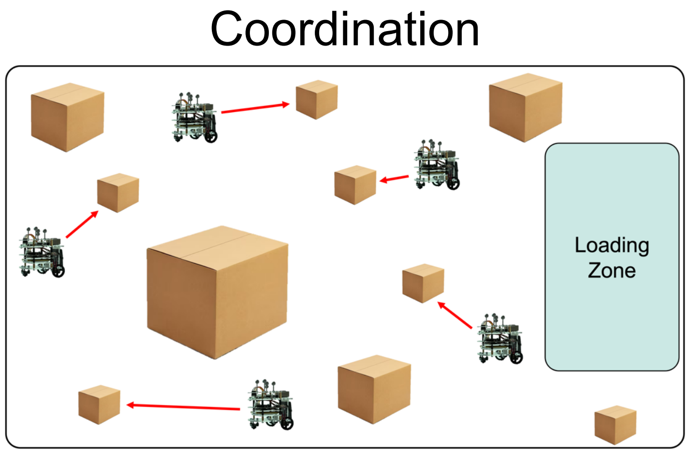
The Avengers fight Loki's army in NYC [1]

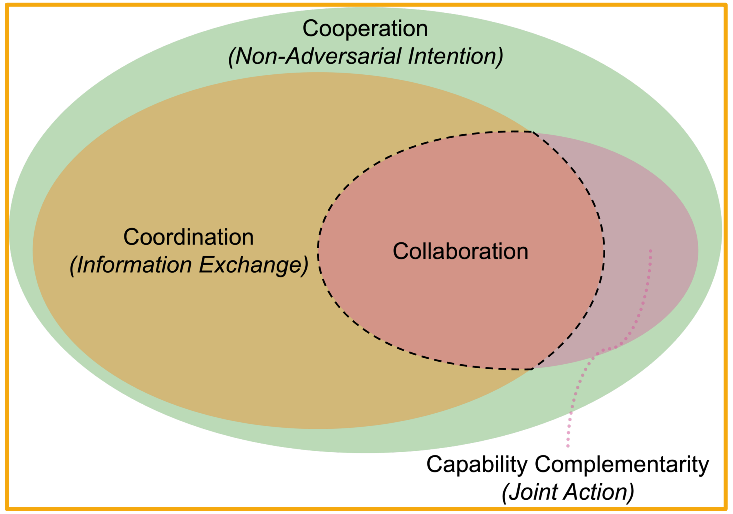
- Multi-hero system
- Each hero brings something to the table (power or ability)
- They work together and combine their abilities to complete their mission

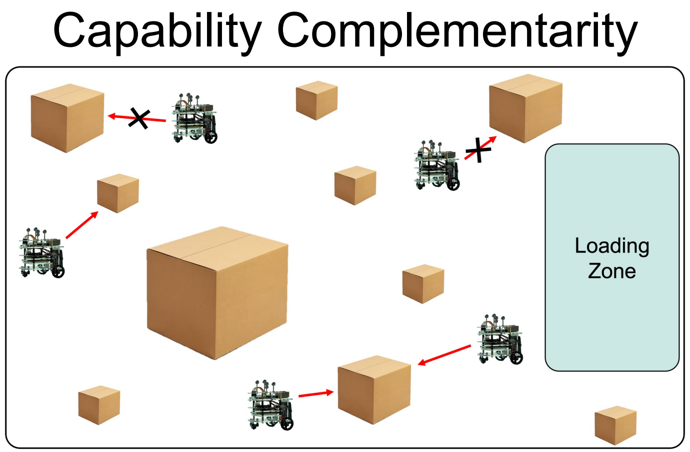
[1]: https://www.polygon.com/22893531/avengers-2012-marvel-movies-phase-4/

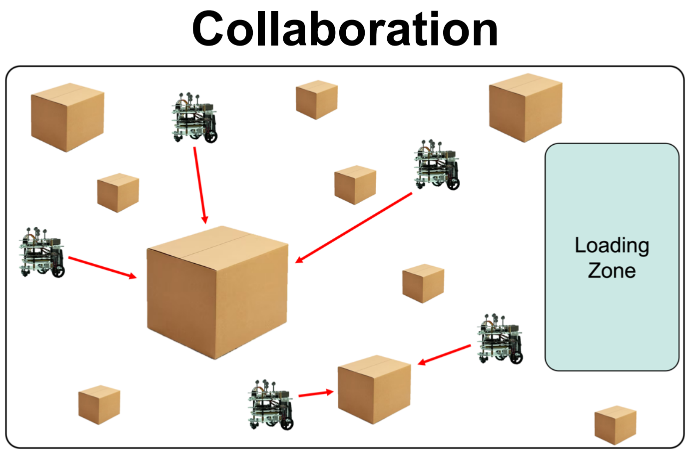
Collaboration in Multi-Agent Systems



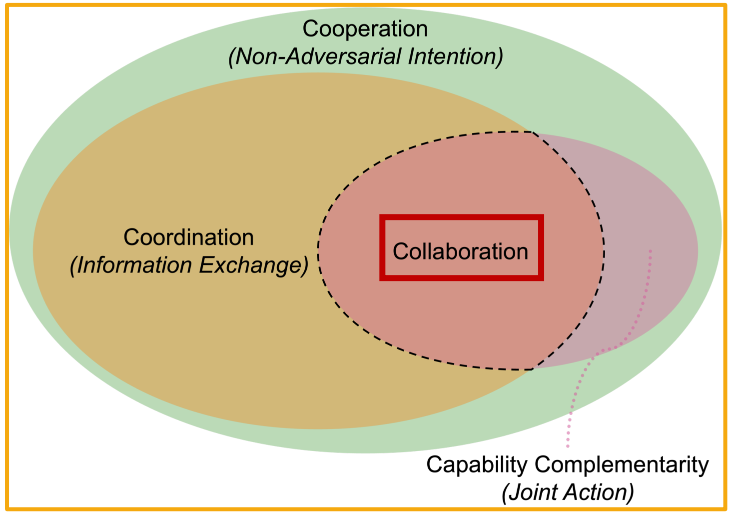


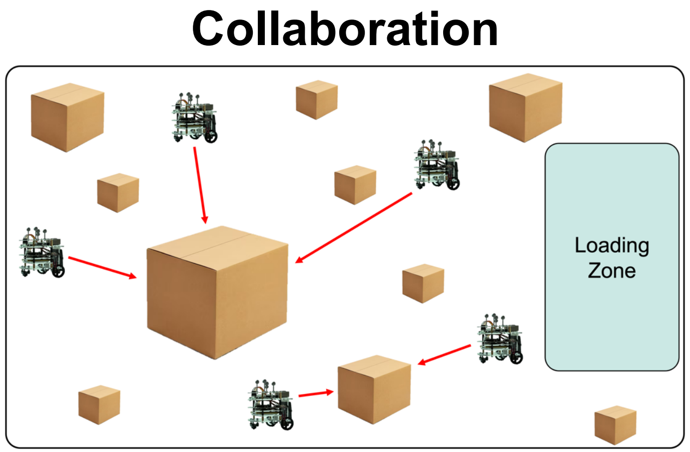






Collaboration in Multi-Agent Systems





Collaboration in Multi-Agent Systems

Papers	Centralized	Decentralized	Hierarchical	Ecological	HSI	Game Theoretic	Learning-Based
[7, 9, 104, 105]	×	✓	×	×	×	×	✓
[8, 57, 77]	✓	×	×	✓	×	×	×
[30]	×*	✓	×	×	×	×	✓
[58, 59]	✓	×	×	×	×	×	×
[60]	✓	×	×	×	×	×	✓
[61, 76]	×	✓	×	✓	×	×	×
[62, 63, 94, 95, 99, 100]	×	✓	×	×	×	✓	×
[64, 65]	×	×	✓	×	×	×	✓
[66]	×	×	✓	×	×	×	×
[67, 68, 89, 90]	×	✓	×	×	✓	×	×
[78]	×	✓	×	✓	×	×	✓
[87, 91]	×	×	✓	×	✓	×	×
[88]	×	×	✓	×	✓	×	✓

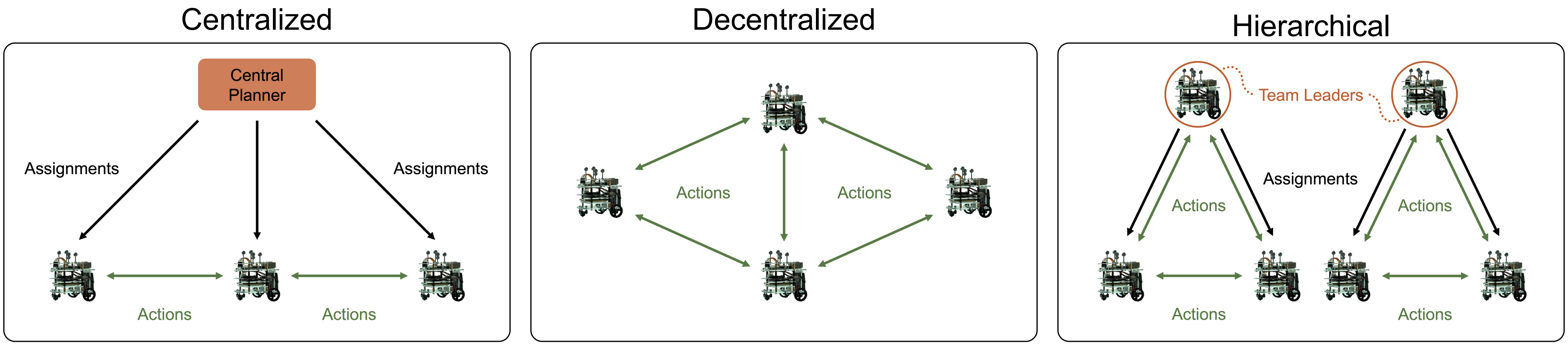
Why decentralized?

- Scalability
- Robustness
- Adaptability

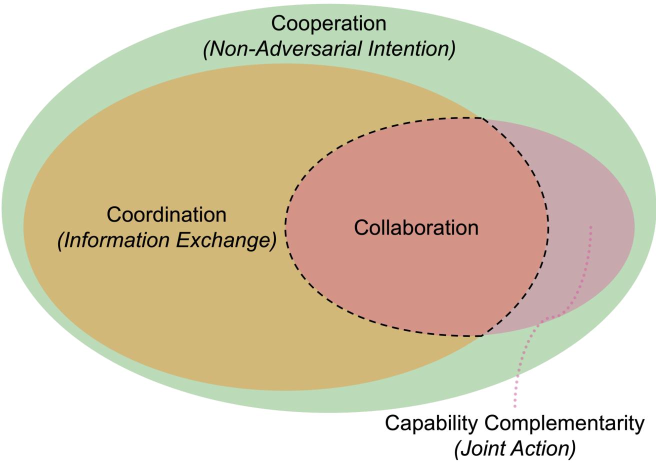
[…]: references cited here are shown as numbered in the corresponding manuscript

Collaboration in Multi-Agent Systems

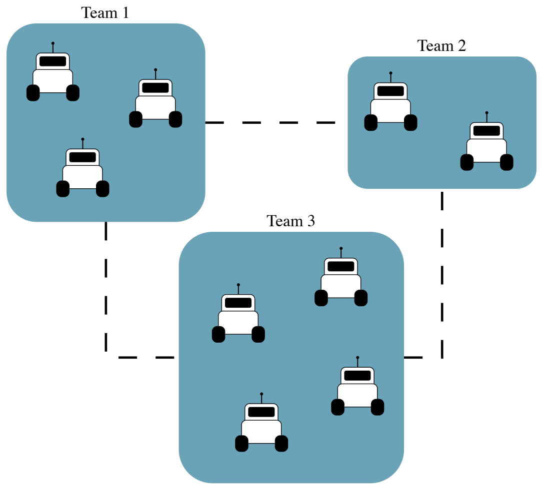
Organizational Structures in Multi-Agent Systems



Collaboration in Multi-Agent Systems [1]



Resource Allocation in Multi-Team Systems [2]



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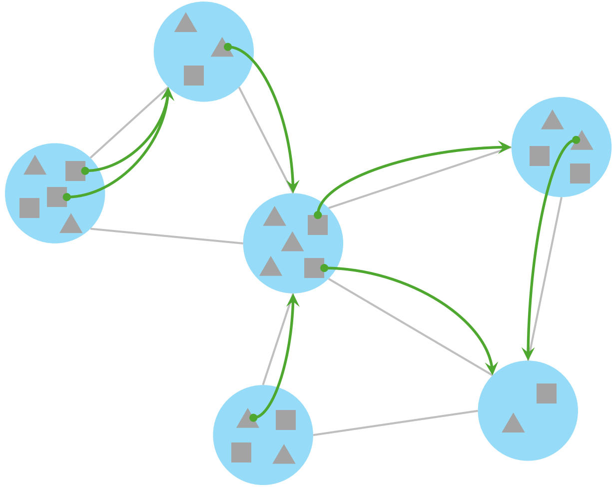

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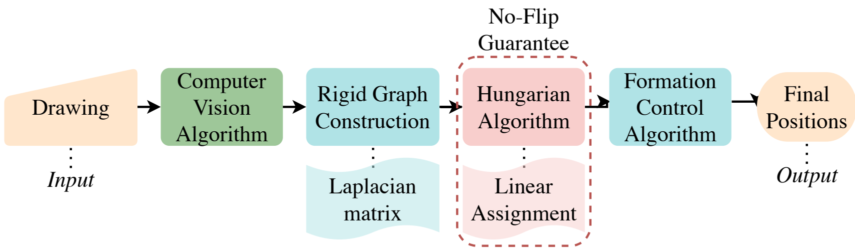

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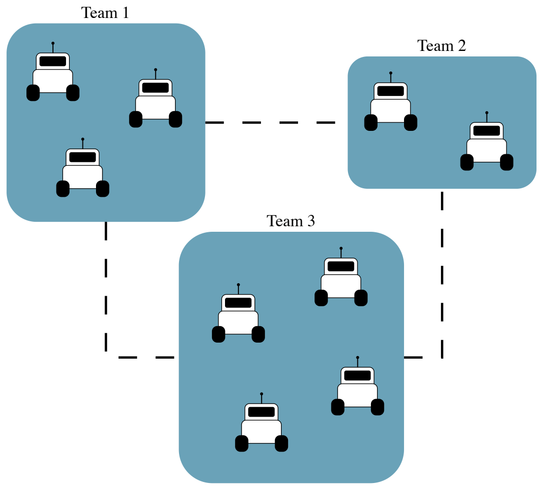
Resource Allocation in Multi-Team Systems [2]



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Resource Allocation in Multi-Team Systems

Los Angeles County fires: Maps of where the wildfires are [1]

Observations:

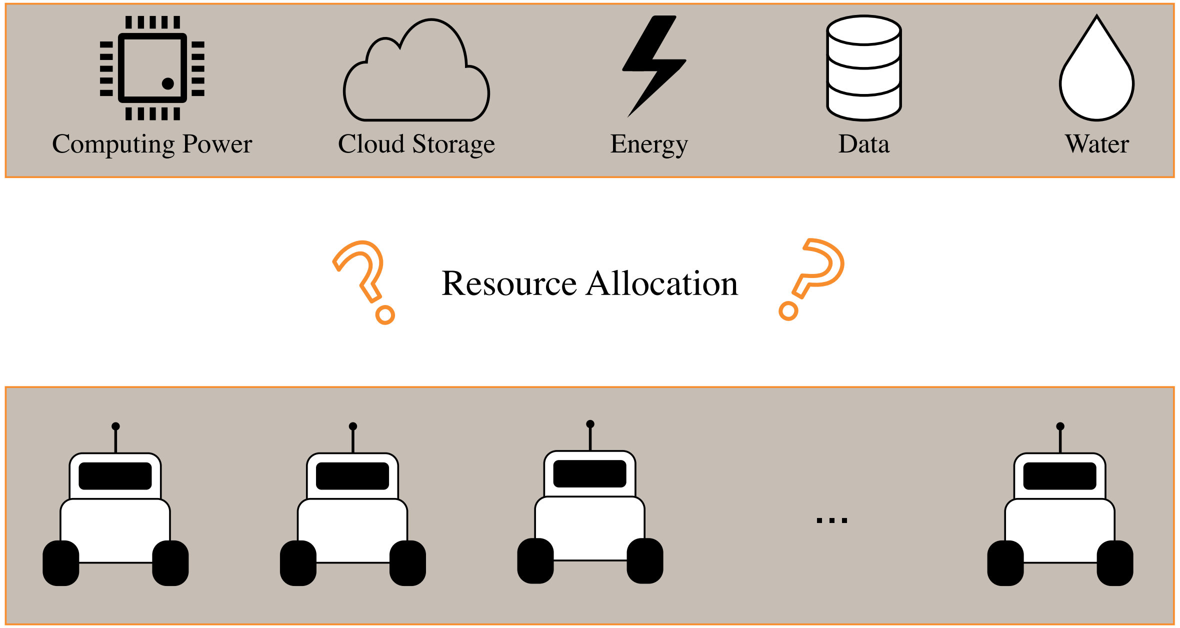
- 4 Distinct Regions of Interest
- Each has a certain fire intensity
- Each has a certain population density

Multi-Robotic Solution?

[1]: https://www.abc10.com/video/news/local/california/los-angeles-county-fires-maps-of-where-the-wildfires-are/103-e4c8ed38-68b4-4e7a-bfca-50073bfbfd3f

Resource Allocation in Multi-Team Systems

Area Coverage with Fire-Fighting Robots [1]



Resource Allocation in Multi-Agent Systems, e.g., [2]

Q1. How to implement resource allocation for such scenarios?

[1]: R. Lin, S. Kim, and M. Egerstedt, “Heterogeneous Collaborative Pursuit via Coverage Control Driven by Fokker–Planck Equations,” IEEE Transactions on Robotics (T-RO), 2025.
[2]: P. Anussornnitisarn, S. Y. Nof, and O. Etzion, “Decentralized control of cooperative and autonomous agents for solving the distributed resource allocation problem,” International Journal of Production Economics, vol. 98, no. 2, pp. 114–128, 2005.

Ecological Inspiration: Decision Making

Lioness adopts deceased sister's cubs [2]

Hamilton's Rule of Altruism: [1]

$$rB \geq C$$

- r : genetic relatedness
- B : benefit to the recipient
- C : cost to the donor

- Genetic relatedness: sibling relationship, lioness and her deceased sister
- Benefit to the lion cubs' deceased mother: her genes will survive through her cubs' survival
- Cost to the lioness: sharing her lion cubs' resources and splitting her care.

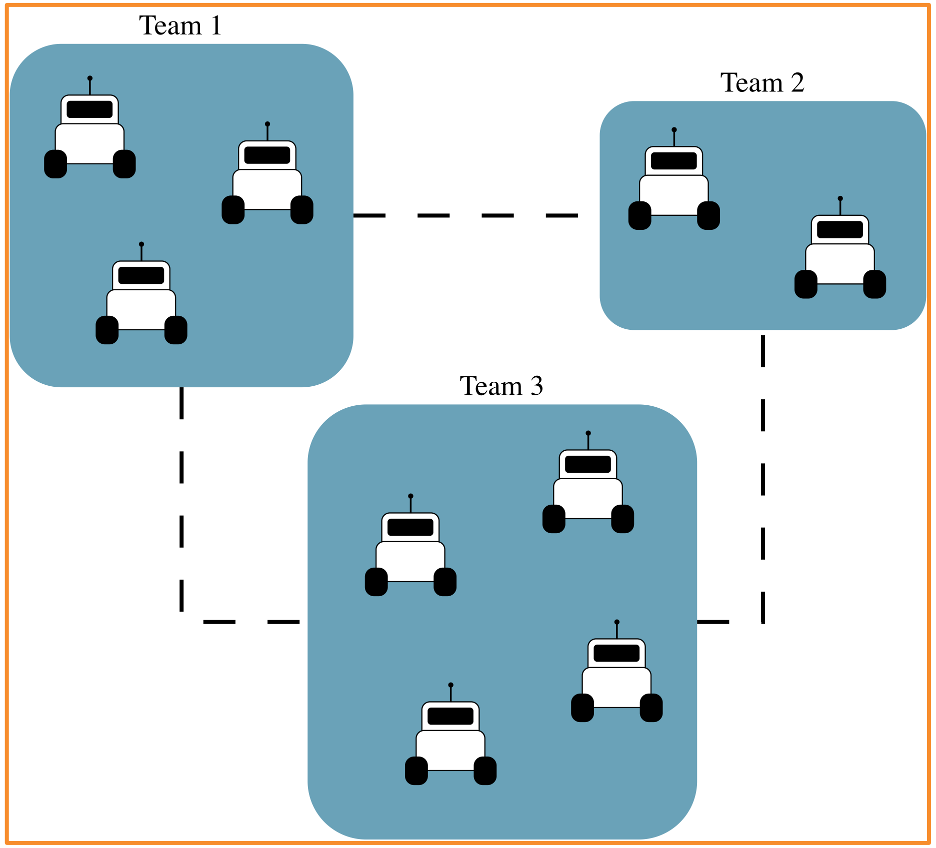
Q2. How to translate Hamilton's rule to robotic systems?

Lioness → **Team of Robots**

Genetic Relatedness → **Mission Weights**

[1]: W. D. Hamilton, "The genetical evolution of social behaviour. ii," Journal of Theoretical Biology, vol. 7, no. 1, pp. 17–52, 1964.
[2]: https://www.dailymail.co.uk/news/article-4418844/Lioness-suckles-orphaned-cubs-adopted-sister.html

Resource Allocation in Multi-Team Systems



Graph Modeling: [1]

- Let there be m teams of robots
- $G = (V, E)$: undirected graph
- $V = \{v_1, \ldots, v_m\}$: set of **teams**
- $E \subseteq V \times V$: set of **potential interactions**

Key Idea:

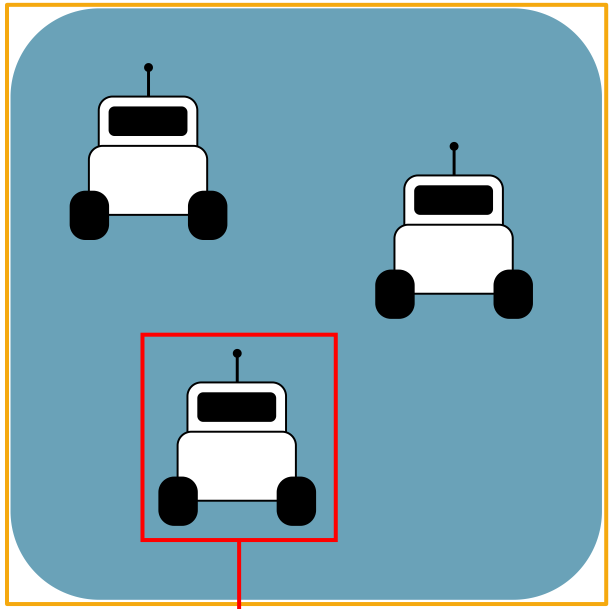
Compute potential interactions inspired by Hamilton's Rule [2]

Assumption 1:

All agents in all m teams are homogenous, i.e., identical

[1]: M. Mesbahi and M. Egerstedt, Graph Theoretic Methods in Multiagent Networks. Princeton University Press, 2010.
[2]: W. D. Hamilton, "The genetical evolution of social behaviour. ii," Journal of Theoretical Biology, vol. 7, no. 1, pp. 17–52, 1964.

Resource Allocation in Multi-Team Systems



Robot = Resource

How to define a team's mission performance?

Mission evaluation function

Which parameter(s) to consider?

Number of robots in a team

Mission Evaluation Function for team k: $F_k(n_k)$, with n_k number of robots in team k

Resource Allocation in Multi-Team Systems

Mission Evaluation Function for team k: $F_k(n_k)$, with n_k: number of robots in team k

Assumption 2:

For each team k, the mission evaluation function $F_k(n_k) : \mathbb{N} \to \mathbb{R}$ satisfies the following properties:

1. Strictly Increasing: For all $n \geq 1$,
$$F_k(n+1) > F_k(n)$$
2. Discrete Concavity: For all $n \geq 2$,
$$F_k(n+1) - F_k(n) \leq F_k(n) - F_k(n-1)$$

Higher is better

Diminishing Returns

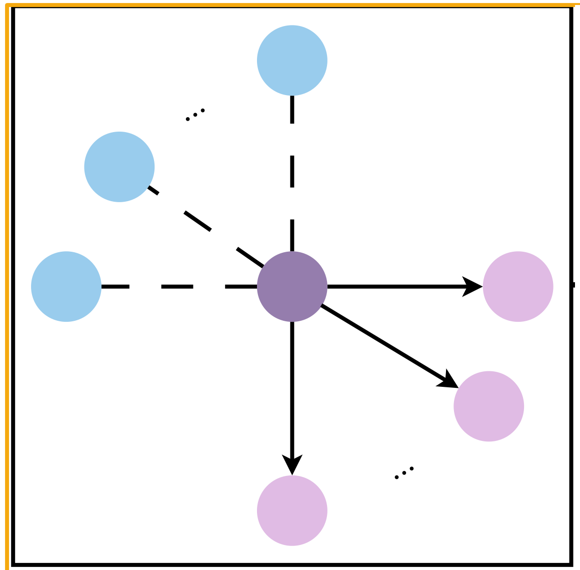
Resource Allocation in Multi-Team Systems

Hamilton's Rule [1]	**Hamilton's Rule for Team-Level Resource Allocation**	
$rB \geq C$	$r_{ij}B_j > C_i, \quad (i,j) \in E$	
r: genetic relatedness	r_{ij}: relative mission importance	$r_{ij} = \dfrac{w_j}{w_i} \quad and \quad r_{ji} = \dfrac{w_i}{w_j} = \dfrac{1}{r_{ji}}$
B: benefit to the recipient	B_j: benefit to team j	$B_k = F_k(n_k+1) - F_k(n_k)$
C: cost to the donor	C_i: cost to team i	$C_k = F_k(n_k) - F_k(n_k-1)$

Theorem 1. Consider two teams i and j, with $i, j \in V$, each with at least one agent. Under Assumptions 1 and 2, if $r_{ij}B_j > C_i$, it follows that $r_{ji}B_i \leq C_j$.

[1]: W. D. Hamilton, "The genetical evolution of social behaviour. ii," Journal of Theoretical Biology, vol. 7, no. 1, pp. 17–52, 1964.

Resource Allocation in Multi-Team Systems



Legend:

Team $k \in V$

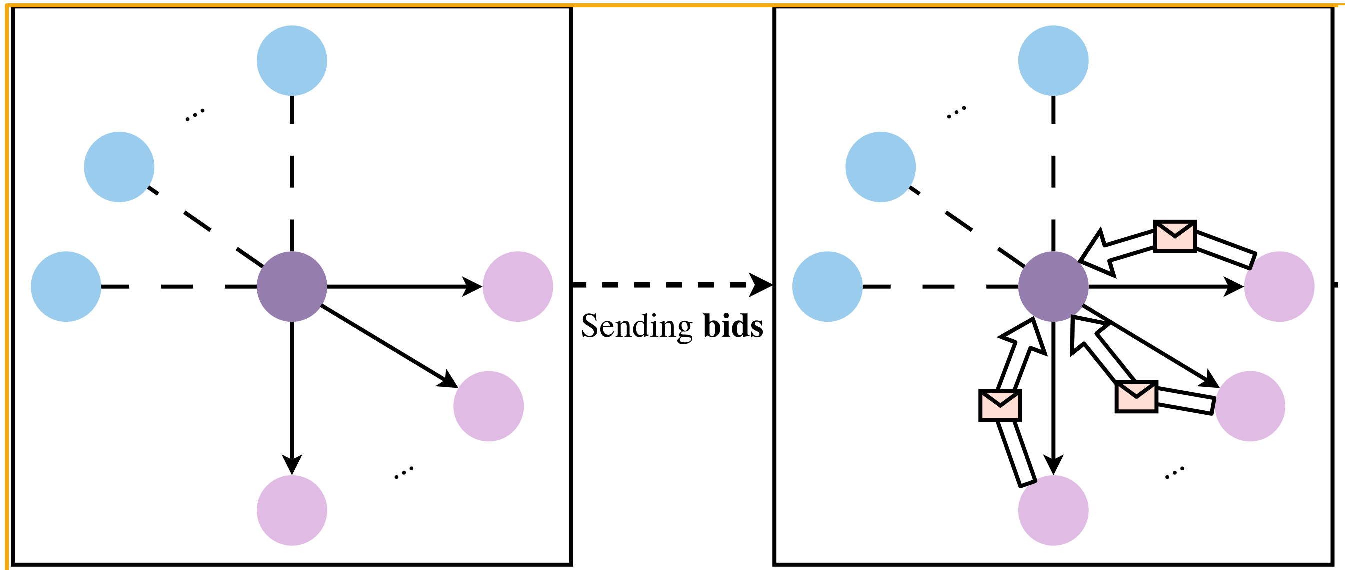
Team $o \in \mathcal{N}_k$

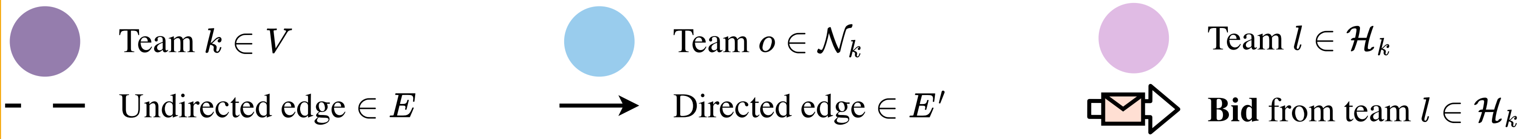
Team $l \in \mathcal{H}_k$

Undirected edge $\in E$

Directed edge $\in E'$

Resource Allocation in Multi-Team Systems





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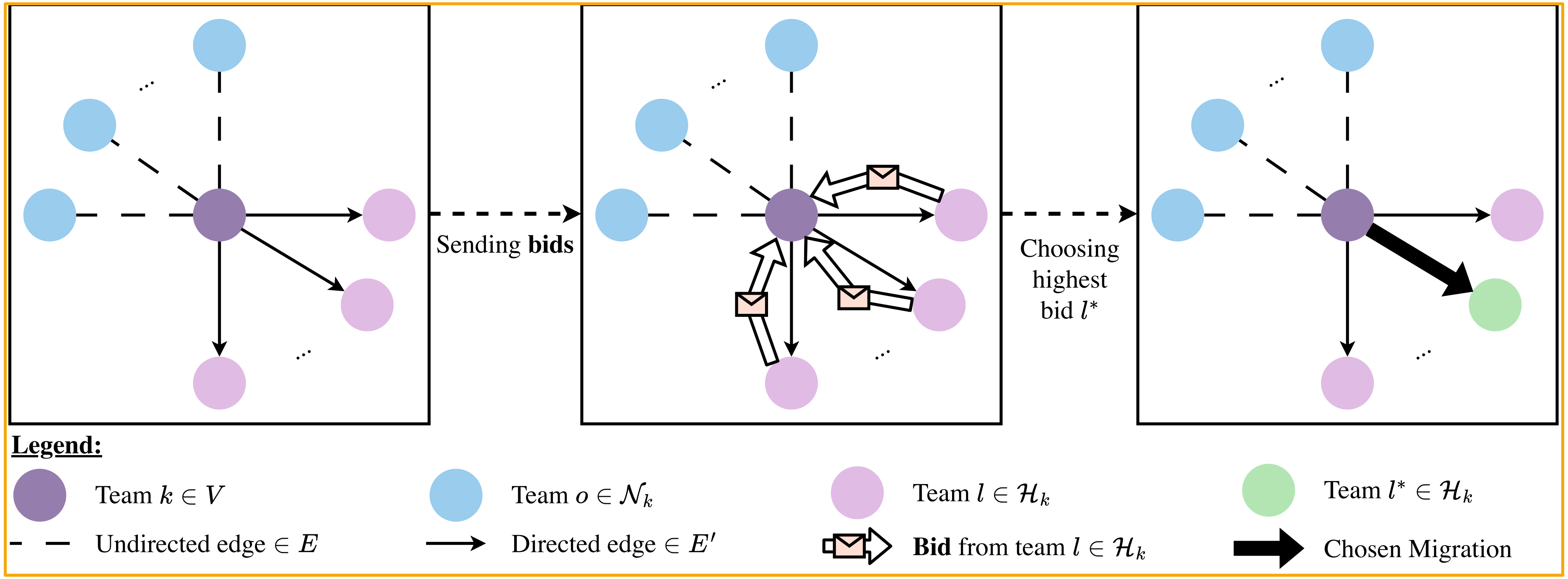
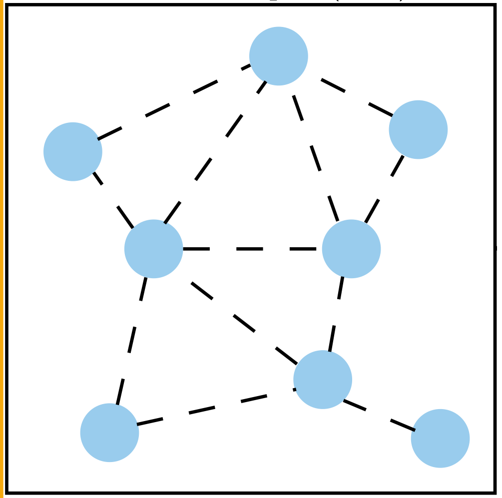
Resource Allocation in Multi-Team Systems



Resource Allocation in Multi-Team Systems

Undirected Graph $G(V, E)$

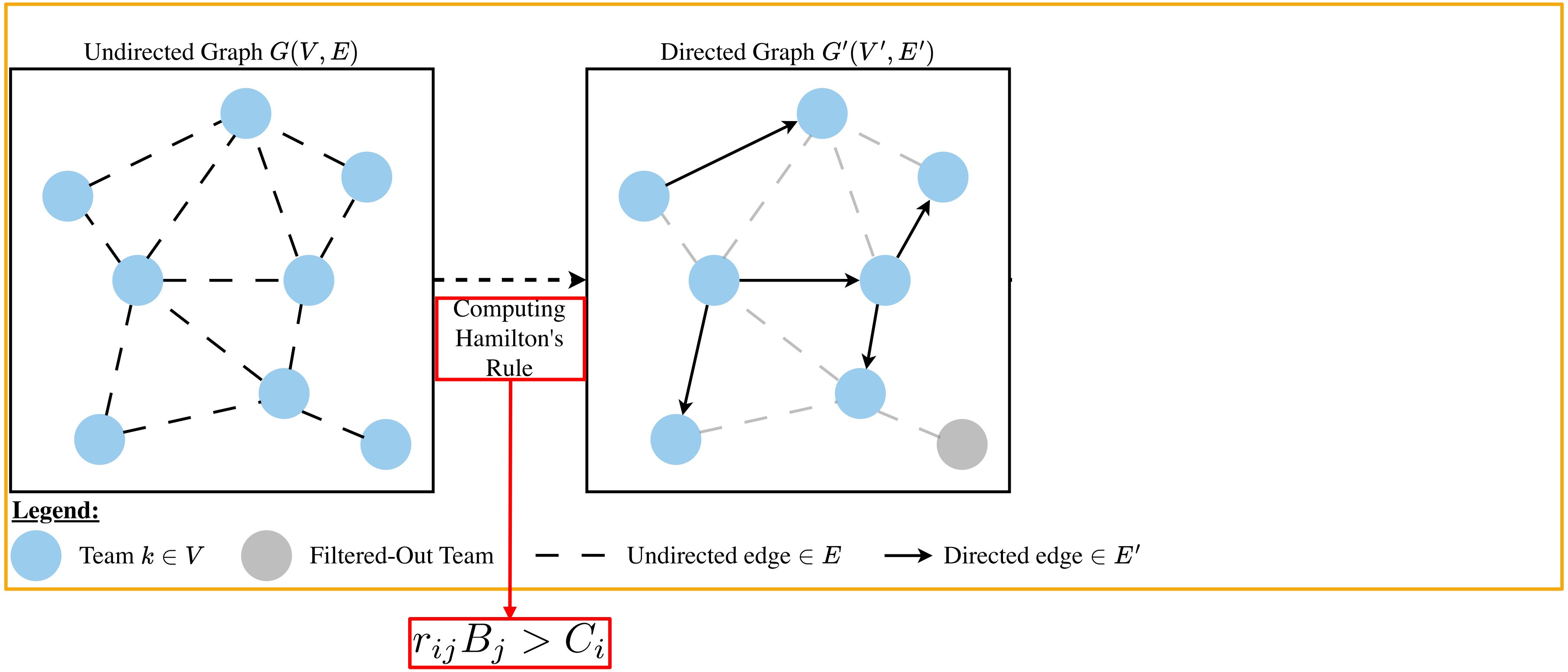


Legend:

Team $k \in V$

Undirected edge $\in E$

Resource Allocation in Multi-Team Systems



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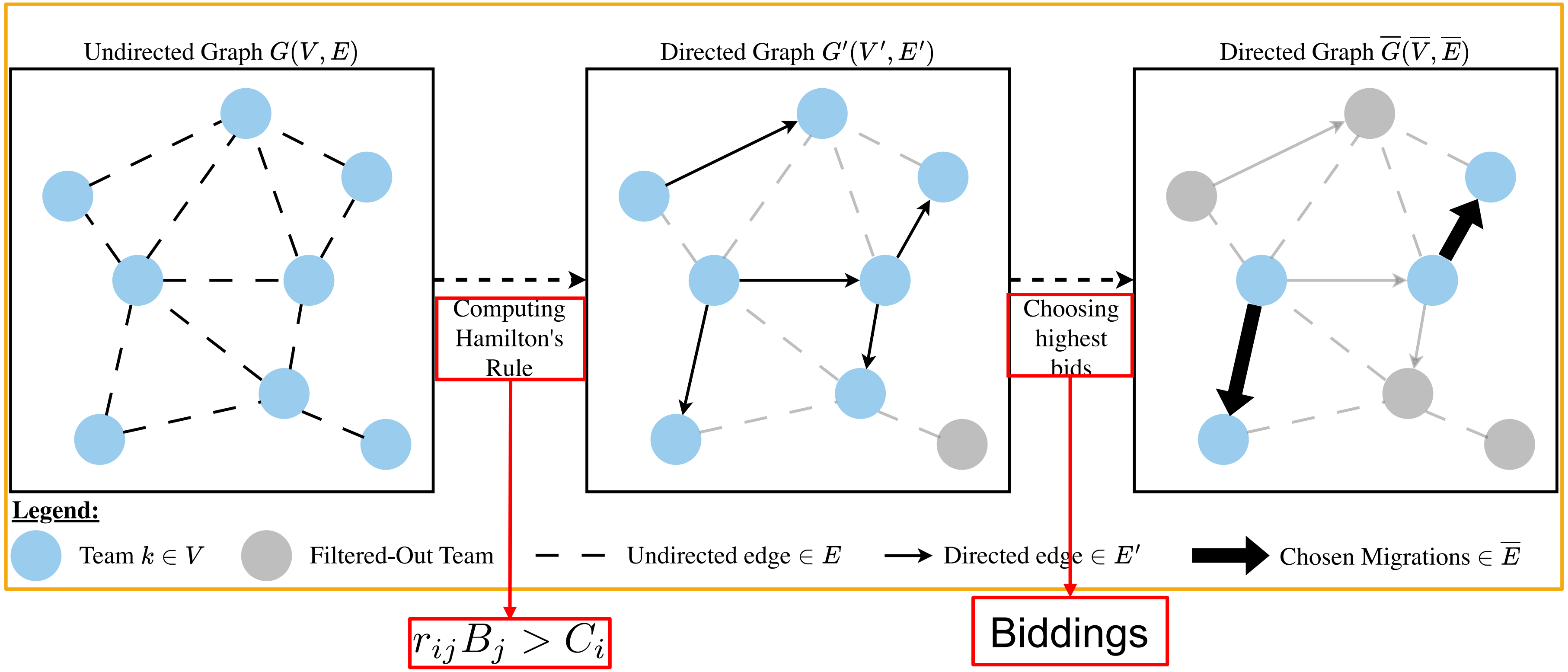

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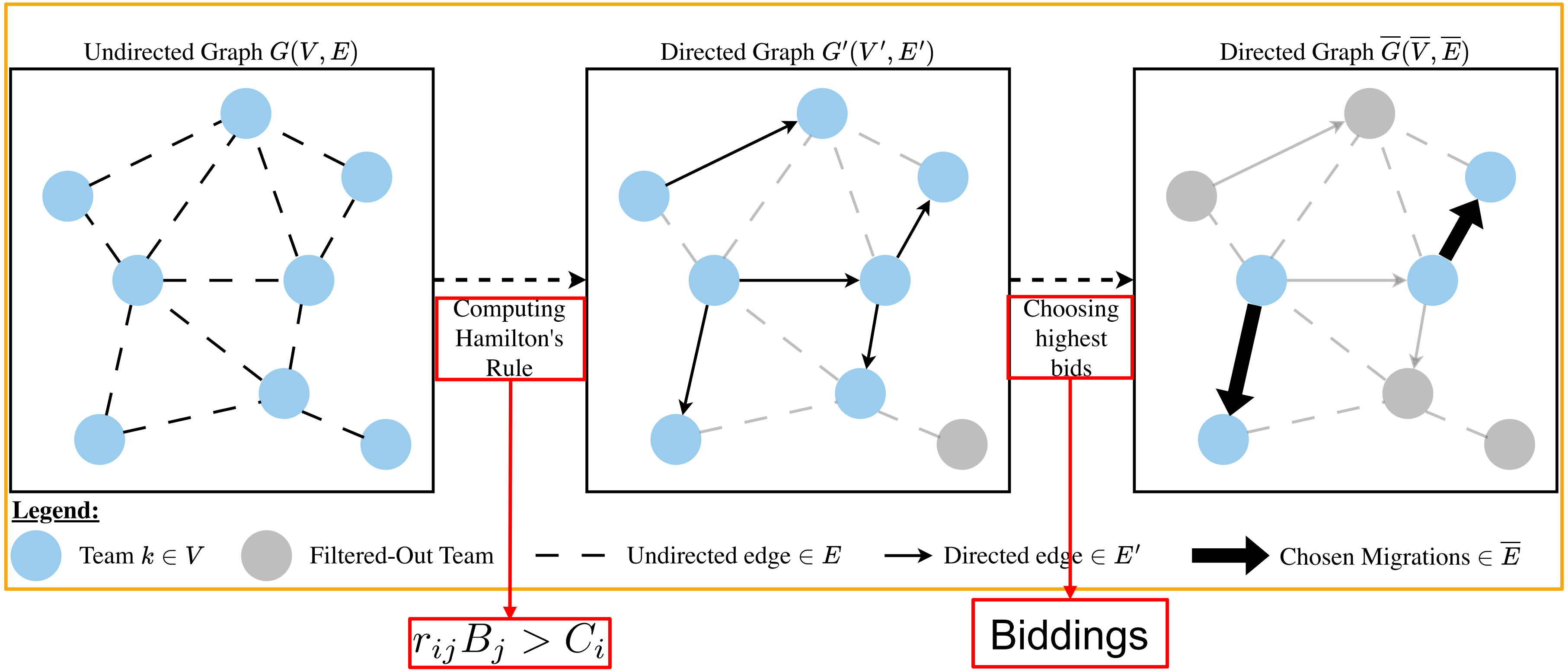


Is this globally optimal?

Resource Allocation in Multi-Team Systems

Global Mission Evaluation Function:

$$\mathcal{G}(n_1, \ldots, n_m) = \sum_{k=1}^{m} w_k \, F_k(n_k)$$

N total number of robots across the m teams

Global Inequality Rule:

$$\mathcal{G}(n'_1, \ldots, n'_m) > \mathcal{G}(n_1, \ldots, n_m)$$

Global benefit instead of local (or greedy)

Philosophy:
"If all of us benefit, each of us benefits"

$\Omega = \left\{ (n_1, \ldots, n_m) \in \mathbb{Z}^m_{>0} \,\middle|\, \sum_{k=1}^{m} n_k = N \right\}$: set of all allocations of the N agents between the m teams

More formally,

$$\mathcal{G} : \Omega \to \mathbb{R}$$

Resource Allocation in Multi-Team Systems

Lemma 1. There exists an allocation $(n_1^*, \ldots, n_m^*) \in \Omega$, such that

$$\mathcal{G}(n_1^*, \ldots, n_m^*) \geq \mathcal{G}(n_1, \ldots, n_m) \quad \forall (n_1, \ldots, n_m) \in \Omega$$

In other words, $\mathcal{G}$ attains a maximum value on Ω, and $(n_1^*, \ldots, n_m^*)$ is an optimal allocation.

Inputs:

- Initial Allocation
- Mission Evaluation Functions
- Team Weights

1. Compute $\mathcal{G}^{(t)}$
2. Construct $G(V, E)$
3. Construct $G'(V', E')$
4. Bidding Process
5. Construct $\overline{G}(\overline{V}, \overline{E})$
6. Compute $\mathcal{G}^{(t+1)}$
7. Global Inequality Rule
8. Update allocation if applicable
9. Stop when $\mathcal{G}^{(t)} \geq \mathcal{G}^{(t+1)}$

Output:
Final Allocation

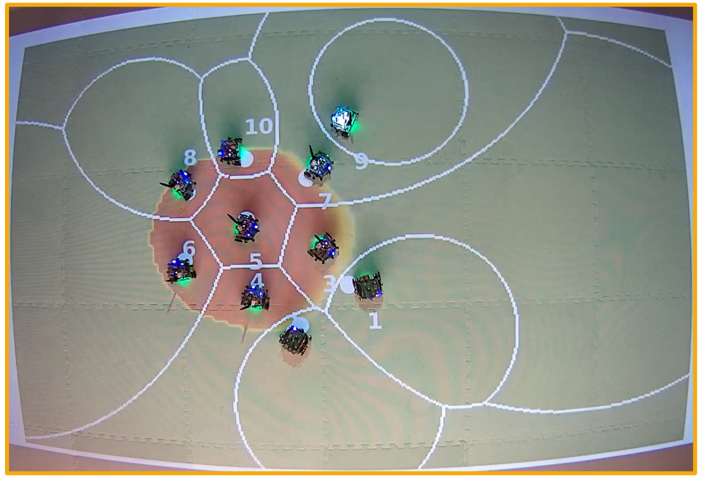
Theorem 2. This iterative process terminates in a finite number of steps and converges to an allocation $(n_1^*, \ldots, n_m^*) \in \Omega$ that maximizes $\mathcal{G}$.

Resource Allocation in Multi-Team Systems

Team Mission → Optimal Area Coverage → Voronoi-Based Coverage Control [1]

$N \in \mathbb{Z}^+$ Robots → Convex and Compact $\mathcal{D} \subset \mathbb{R}^d$ →

$$L(p) = \sum_{i \in \mathcal{N}} \int_{\mathcal{V}_i} \|p_i - q\|^2 \phi(q)\, dq$$



Dynamic Area Coverage with Fire-Fighting Robots [2]

Centroidal Voronoi Tessellation (CVT)

$$p_i = c_i = \frac{\int_{\mathcal{V}_i} q\phi(q)\, dq}{\int_{\mathcal{V}_i} \phi(q)\, dq}, \quad \forall i \in \mathcal{N}$$

- Centroid: $c_i \in \mathbb{R}^d$
- CVT: $c = [c_1^\top, c_2^\top, \ldots, c_N^\top]^\top$

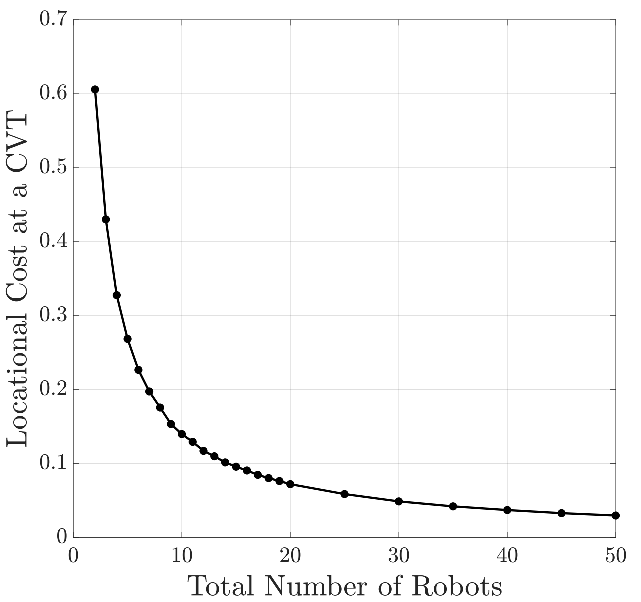
[1] J. Cortes, S. Martinez, T. Karatas, and F. Bullo, "Coverage control for mobile sensing networks," IEEE Transactions on Robotics and Automation, vol. 20, no. 2, pp. 243–255, 2004.
[2]: R. Lin, S. Kim, and M. Egerstedt, "Heterogeneous Collaborative Pursuit via Coverage Control Driven by Fokker–Planck Equations," IEEE Transactions on Robotics (T-RO), 2025.

Resource Allocation in Multi-Team Systems

Mission Evaluation Function: $F(N) = -L(N, c)$, with: $L(N, p) = \sum_{i \in \mathcal{N}} \int_{\mathcal{V}_i} \|p_i - q\|^2 \phi(q)\, dq$

Assumption 2:
Mission Evaluation Function Properties

Discrete Convexity:



Strictly Decreasing:

Theorem 3. Suppose $p_i \in \mathcal{D}$, $\forall i \in \mathcal{N}$. By adding one more robot $p_{N+1} \in \mathcal{D} \setminus \{p_i\}$, $\forall i \in \mathcal{N}$, the locational cost decreases, i.e.,

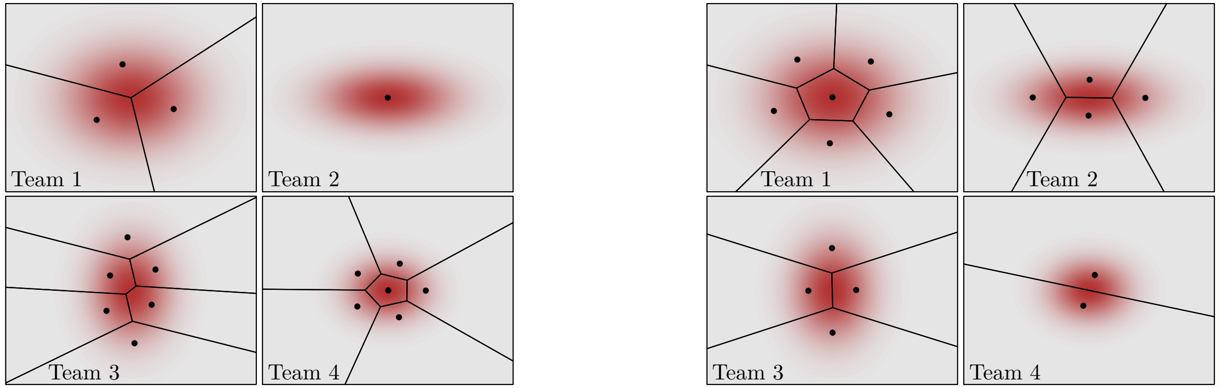
$$L(N+1,\, p') < L(N,\, p), \quad \forall N \in \mathbb{Z}^+$$

where $p' = [p^\top, p_{N+1}^\top]^\top$.

Resource Allocation in Multi-Team Systems

Case 1:

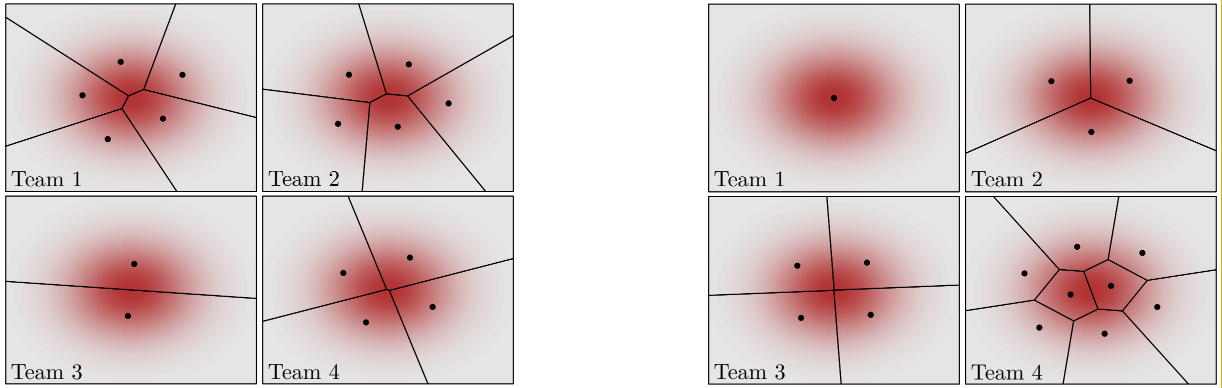
- **Same** mission importances
- **Different** density functions



Initial Allocation Final Allocation

Case 2:

- **Different** mission importances
- **Same** density functions



Resource Allocation in Multi-Team Systems

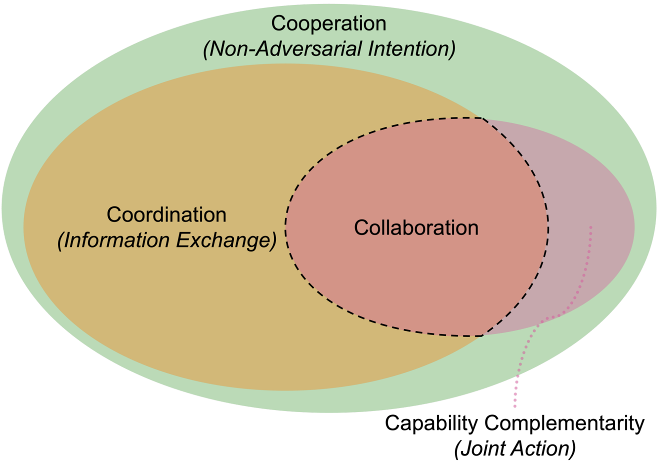
- Ecologically-Inspired Framework: Hamilton's Rule
- Multi-Team Environments
- Optimal Resource Allocation
- Robots Considered as Resources

Framework Achieves Optimal Performance for the Entire System through Altruistic Behavior

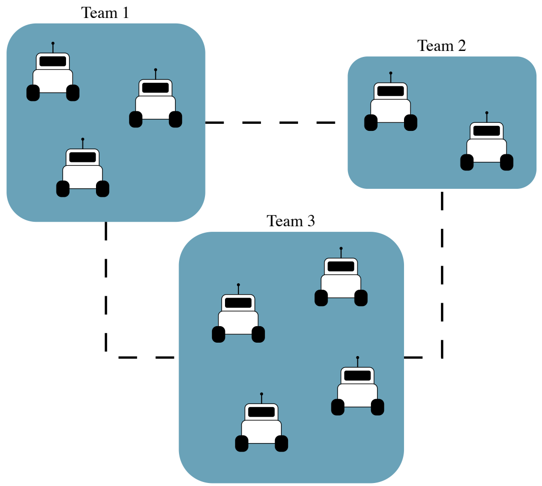
BUT!

- Assumptions on $F_k(n_k)$
- Homogeneous robots
- No migration cost
- Centralized
- Average case complexity: $O(|V||E|)$
- Worst case complexity: $O\left(|E|\binom{N-1}{M-1}\right)$
- And what about *collaboration*?

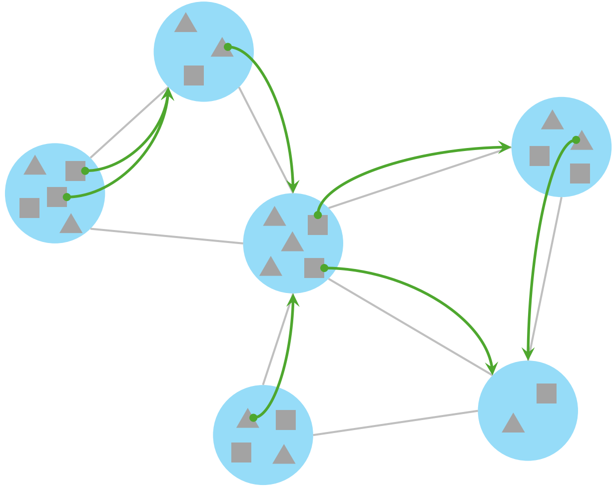
Collaboration in Multi-Agent Systems [1]



Resource Allocation in Multi-Team Systems [2]

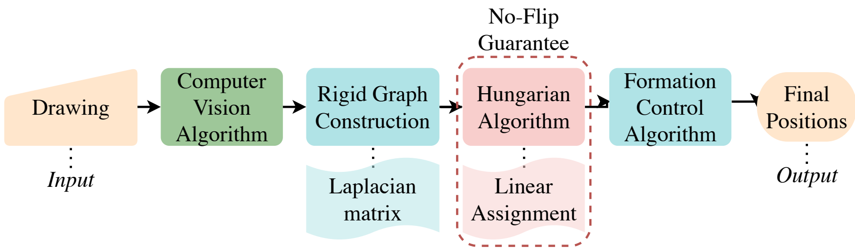


Learning Heterogeneous Altruistic Collaboration [3]



Learning Collaborative Robot Mobility with Model Predictive Scheduling

Human-Swarm Collaboration for Free Multi-Robot Formations



[1]: **Karam, R.**, Nguyen, A. A., Lin, R., Martin, D. R., Morales, D., Butler, B. A., & Egerstedt, M., “Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration.” (submitted to CSM, 2026)

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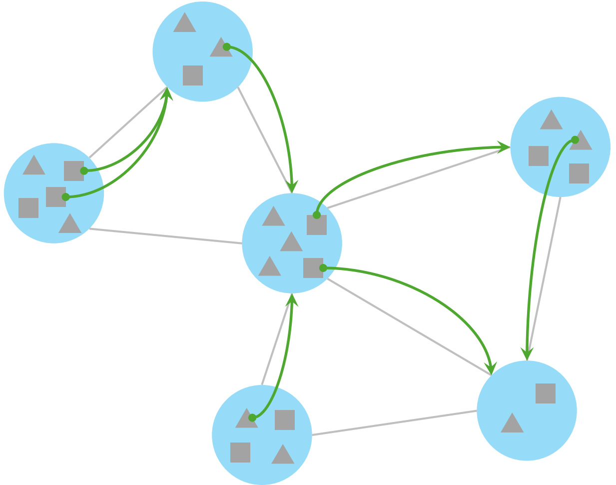
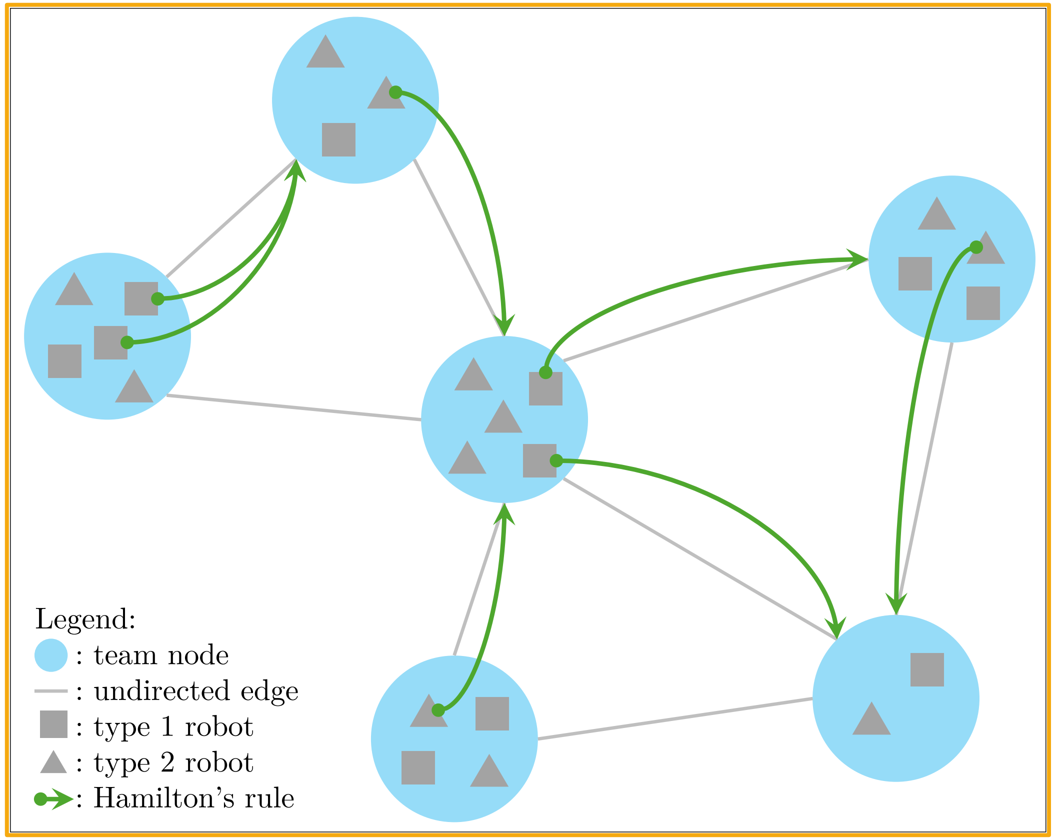


[3]: **Karam, R.**, Lin, R., Butler, B. A., & Egerstedt, M., “Learning Altruistic Collaboration in Heterogeneous Multi-Team Systems.” (submitted to T-RO, 2026)

Learning Heterogeneous Altruistic Collaboration

Kenyan lioness adopts 6 Beisa Oryxes:
A heterogeneous altruism [1]



[1]: https://www.otusafaris.com/kenyan-lioness-story

Learning Heterogeneous Altruistic Collaboration

Heterogeneous Set	→	$\mathcal{S}_v = \{r \in \mathcal{R} \mid x_{r,v} = 1\},\ \mathcal{S}_v \subset \mathcal{R}.$
Heterogeneous Capabilities	→	$\kappa_r \in \{0,1\}^Q$
Mission Evaluation Function	→	$\mathcal{F}_v : \{\mathcal{S} \mid \mathcal{S} \subseteq \mathcal{R}\} \to \mathbb{R}$

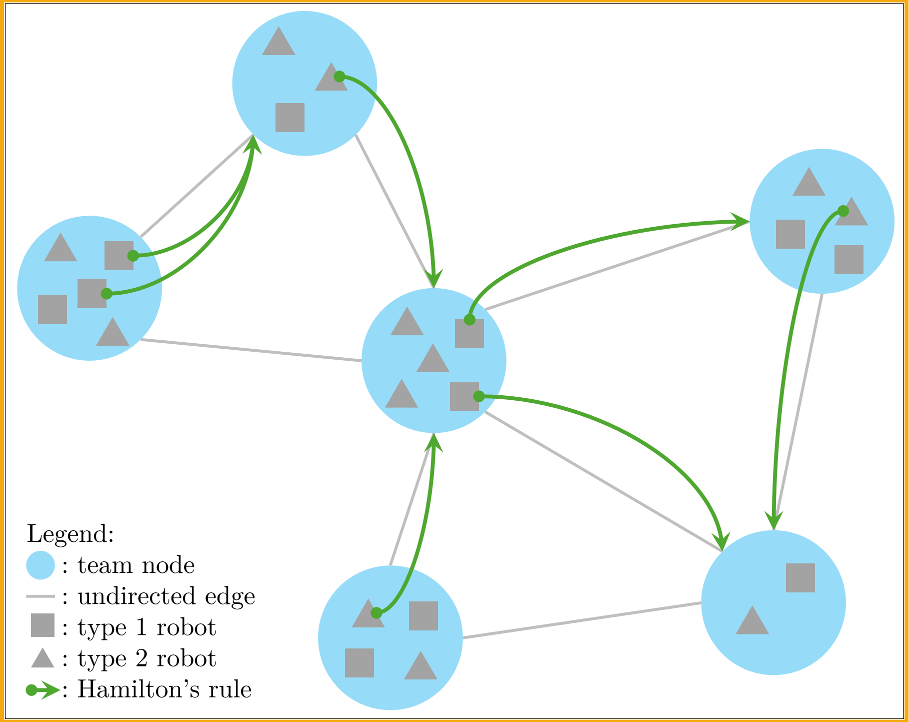
$$\begin{aligned} \max \quad & \text{global performance} \\ \text{s.t.} \quad & \text{assignment constraints} \\ & \text{mission constraints} \\ & \dots \end{aligned}$$

→

$$\begin{aligned} \max \quad & \text{global performance - } \boxed{\text{global cost}} \\ \text{s.t.} \quad & \text{assignment constraints} \\ & \text{mission constraints} \\ & \dots \end{aligned}$$

Learning Heterogeneous Altruistic Collaboration

Hamilton's Rule [1]	Hamilton's Rule for Team-Level Heterogeneous Resource Allocation
$rB \geq C$	$r_{ij}\, B_{r,j}(\mathcal{S}_j) > C_{r,i}(\mathcal{S}_i)$
r	$r_{ij} = \frac{w_j}{w_i} \quad and \quad r_{ji} = \frac{w_i}{w_j} = \frac{1}{r_{ji}}$
B	$B_{r,j}(\mathcal{S}_j) = \mathcal{F}_j(\mathcal{S}_j \cup \{r\}) - \mathcal{F}_j(\mathcal{S}_j)$
C	$C_{r,i}(\mathcal{S}_i) = \mathcal{F}_i(\mathcal{S}_i) - \mathcal{F}_i(\mathcal{S}_i \setminus \{r\})$



The heterogeneous altruistic resource allocation problem is shown to be an NP-Hard problem

[1]: W. D. Hamilton, "The genetical evolution of social behaviour. ii," Journal of Theoretical Biology, vol. 7, no. 1, pp. 17–52, 1964.

Learning Heterogeneous Altruistic Collaboration

$$\begin{aligned} \max \quad & \text{global performance - global cost} \\ \text{s.t.} \quad & \text{assignment constraints} \\ & \text{mission constraints} \\ & \dots \end{aligned}$$

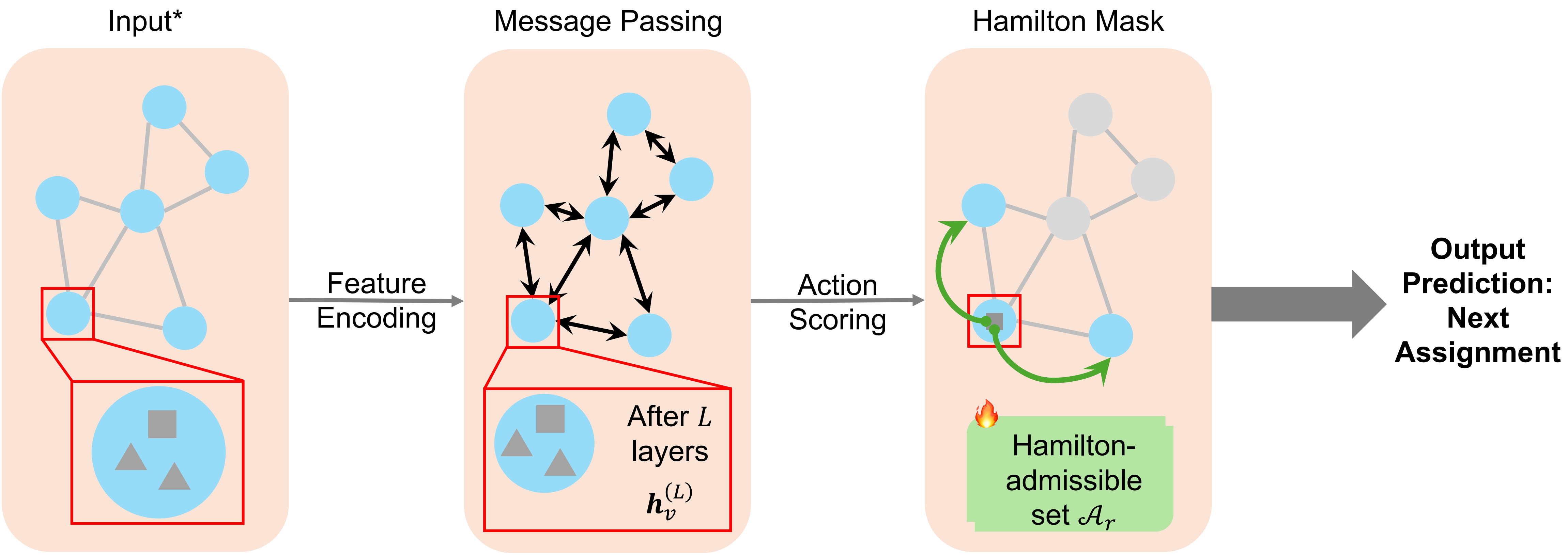
$$\begin{aligned} \max_{X^{\boxed{k+1}}} \quad & \mathcal{G}(X^{\boxed{k+1}}) - \lambda\, \mathcal{C}(X^{\boxed{k}}, X^{\boxed{k+1}}) \\ \text{s.t.} \quad & \text{assignment constraints,} \\ & \boxed{\text{Hamilton's rule constraints}} \\ & \text{mission-specific constraints,} \\ & \dots \end{aligned}$$

Tractability and scalability problem: Solving an NP-hard problem at each iteration of the algorithm!

Learning-Based Solution?

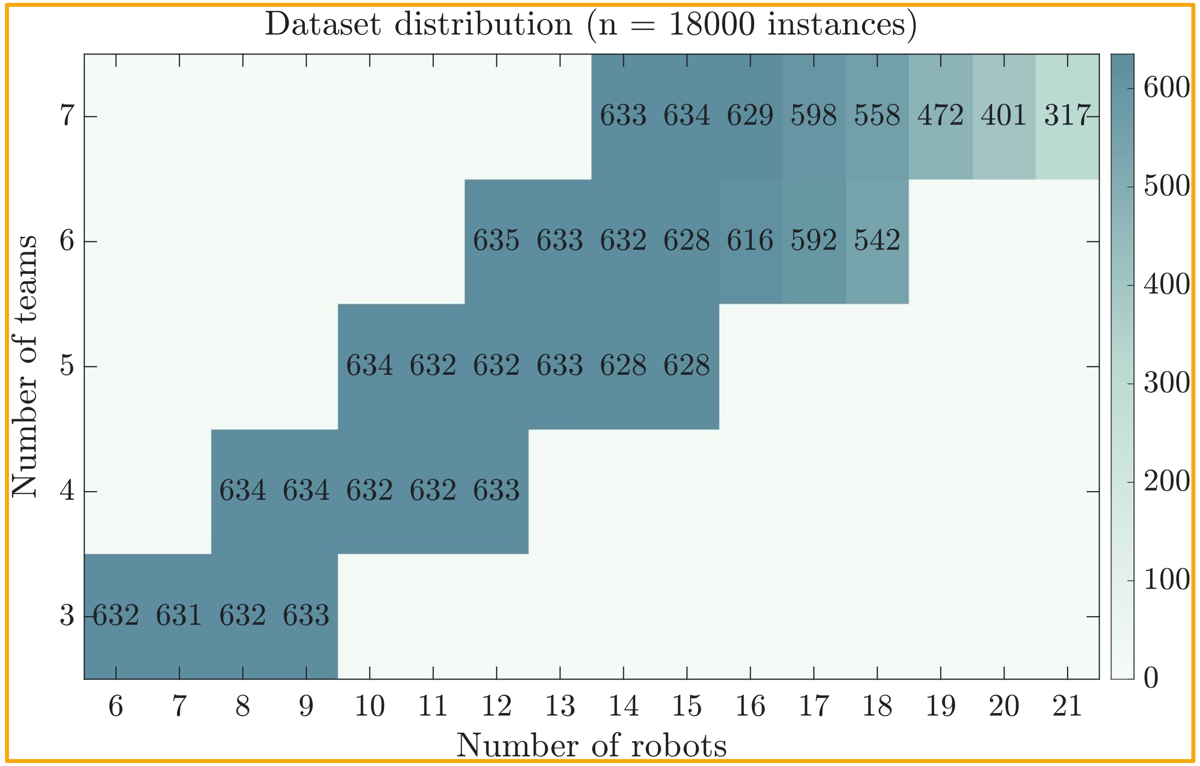
Graph Neural Network policy to approximate the solution of the problem at each iteration

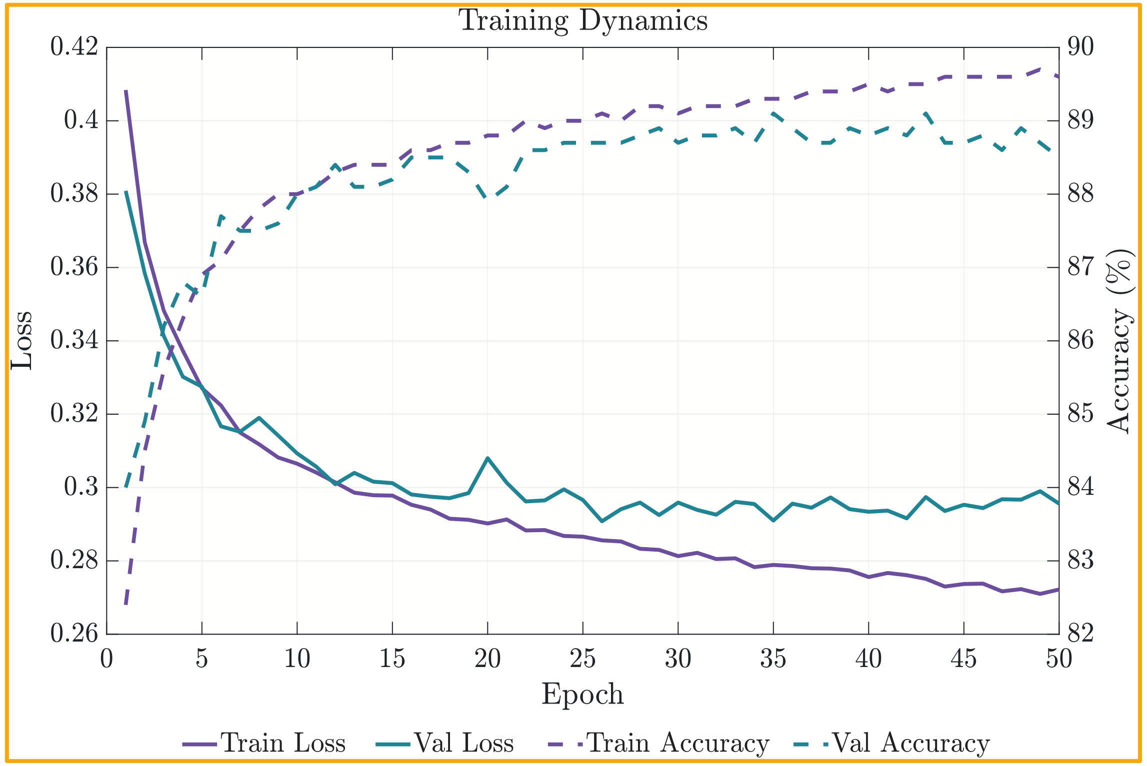
Learning Heterogeneous Altruistic Collaboration



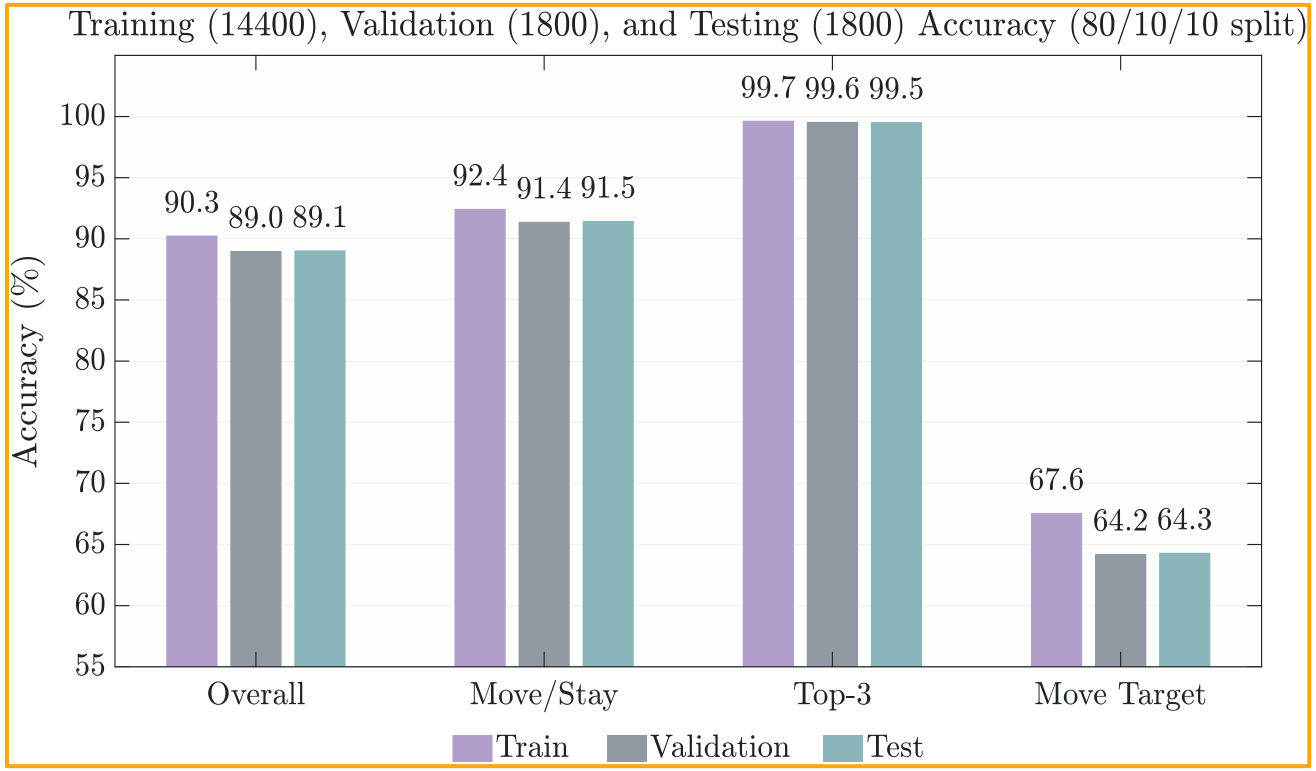
*Team, robot and edge features

Learning Heterogeneous Altruistic Collaboration





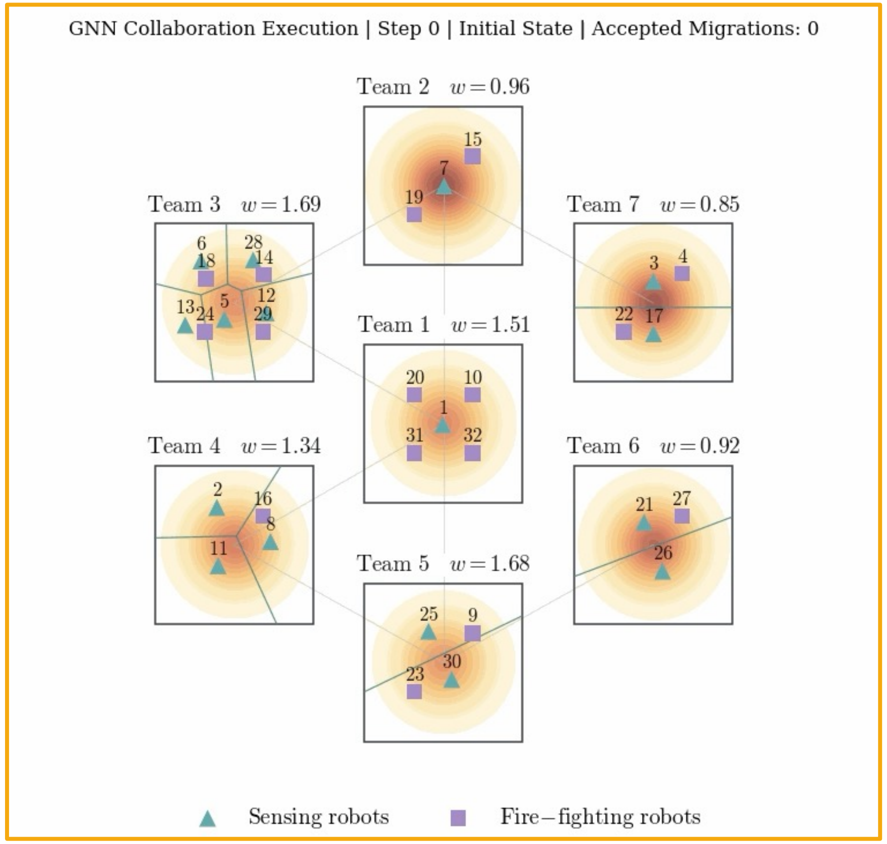
Learning Heterogeneous Altruistic Collaboration

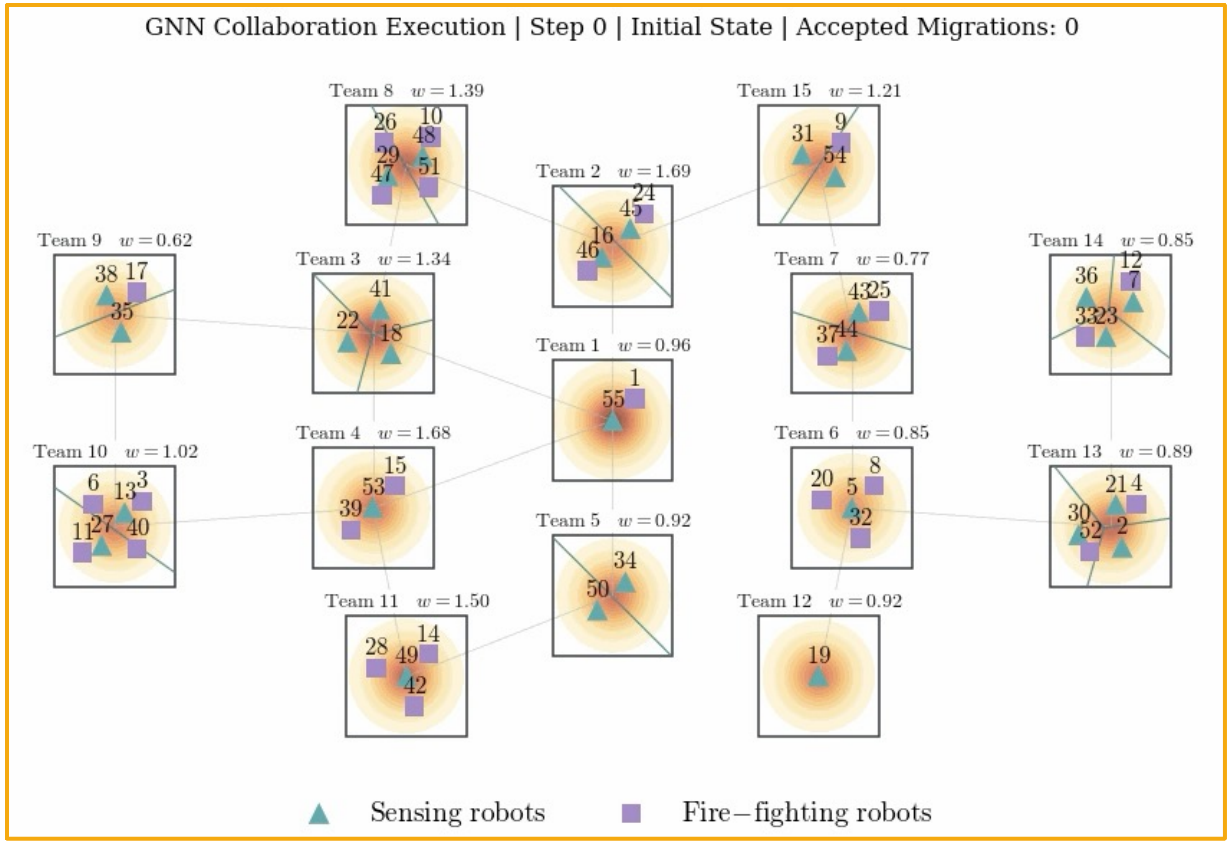


MULTI-SEED ROBUSTNESS

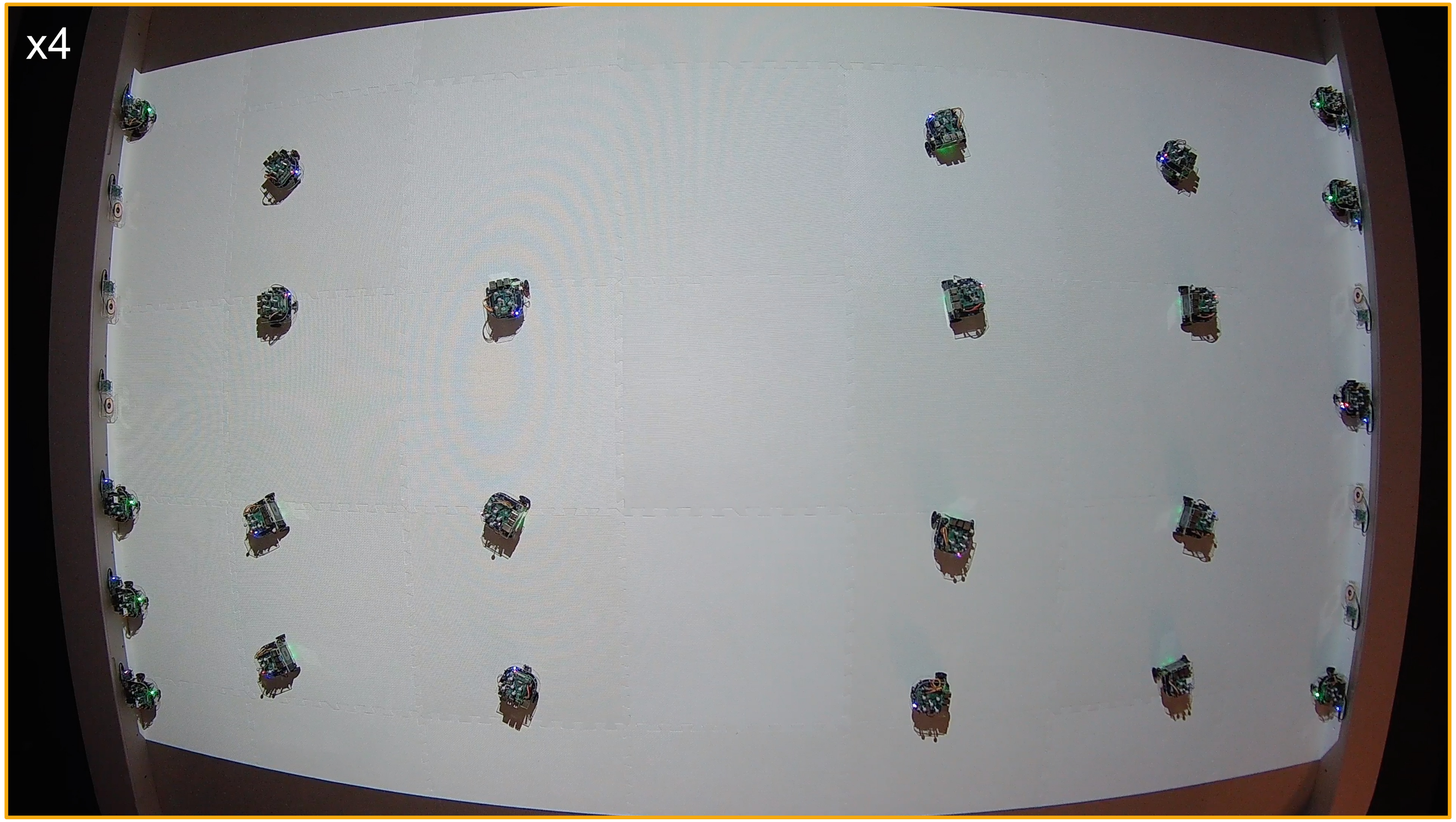
Metric	Mean	Std.	Min	Max
Overall accuracy	88.94%	0.08%	88.79%	89.05%
Move/stay accuracy	91.33%	0.08%	91.22%	91.46%
Top-3 accuracy	99.51%	0.03%	99.46%	99.55%
Move target accuracy	63.29%	1.30%	60.58%	64.33%
Mean loss	0.2361	0.0013	0.2341	0.2376

Learning Heterogeneous Altruistic Collaboration





Learning Heterogeneous Altruistic Collaboration



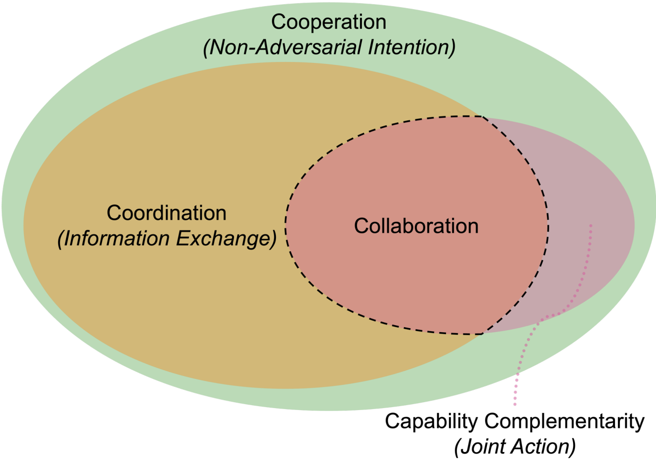
Learning Heterogeneous Altruistic Collaboration

RUNTIME SCALING OF THE EXACT OPTIMIZATION AND THE GNN POLICY. EXACT AND GNN TOTAL RUNTIMES ARE FULL ITERATIVE EXECUTION RUNTIMES. MEAN STEP RUNTIMES ARE COMPUTED FROM THE PER-STEP RUNTIME EXECUTION.

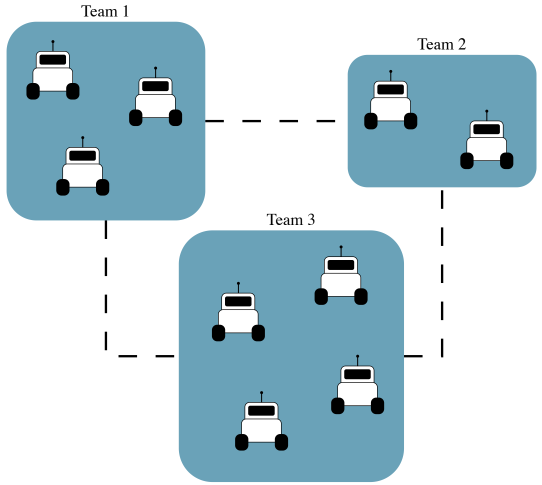
Teams	Robots	Opt. Total (s)	Opt. Mean Step (s)	Opt. Steps	GNN Total (s)	GNN Mean Step (s)	GNN Steps
3	9	0.051	0.025	2	0.048	0.012	4
4	12	0.090	0.030	3	0.012	0.006	2
5	15	0.781	0.390	2	0.009	0.005	2
6	18	35.4	11.8	3	0.170	0.034	5
7	21	∞	–	1	0.201	0.050	4
8	24	∞	–	–	0.298	0.075	4
9	27	∞	–	–	0.816	0.204	4
10	30	∞	–	–	0.658	0.219	3
15	45	∞	–	–	1.957	0.326	6
25	75	∞	–	–	2.787	0.697	4
35	105	∞	–	–	16.3	2.335	7
40	120	∞	–	–	36.0	2.001	18
45	135	∞	–	–	48.5	2.425	20
50	150	∞	–	–	55.3	2.767	20

Concluding Remarks

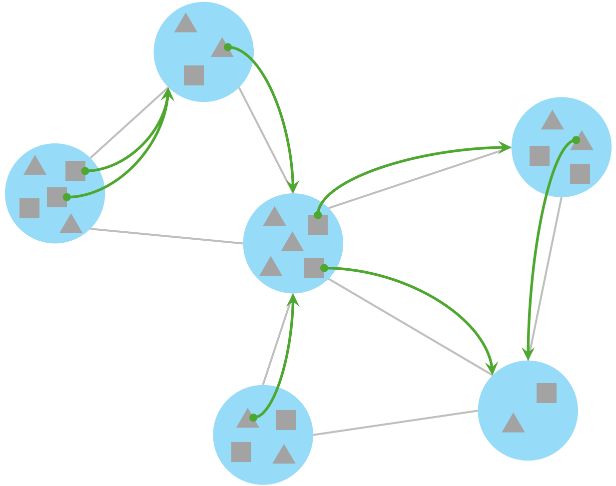
Collaboration in Multi-Agent Systems



Resource Allocation in Multi-Team Systems



Learning Altruistic Collaboration



Collaboration?

- Allows for joint action
- Expands the collective action and feasible sets

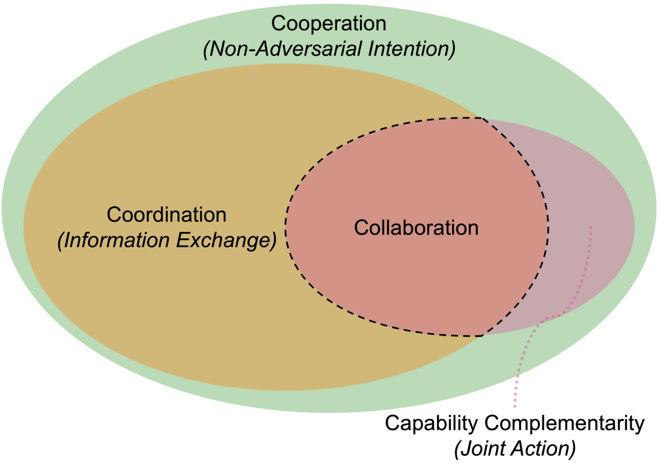
Resource assignment?

- Multi-team setting with robots as resources
- Altruistic decision-making

Learning-based methods?

- Real-time approximation of the intractable NP-hard problem

Collaboration in Multi-Agent Systems [1]



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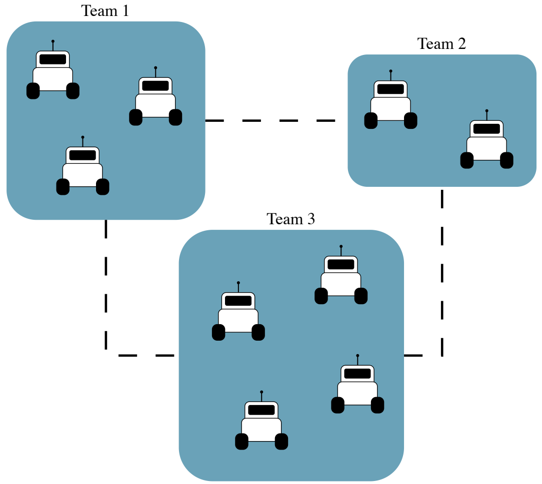

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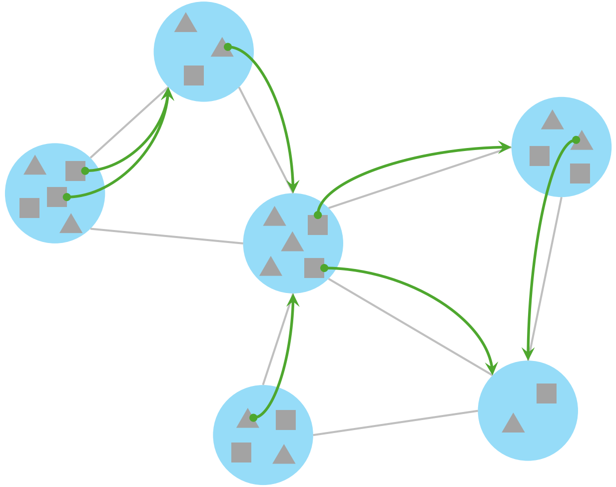
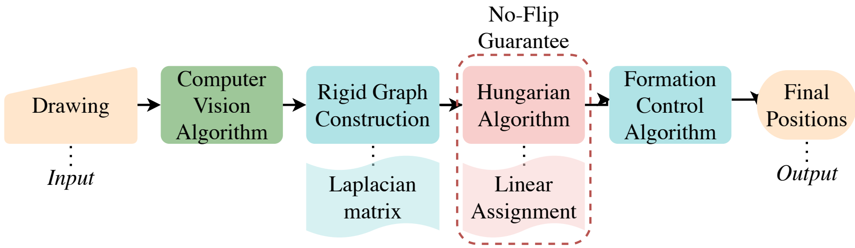


Learning Collaborative Robot Mobility with Model Predictive Scheduling

Human-Swarm Collaboration for Free Multi-Robot Formations



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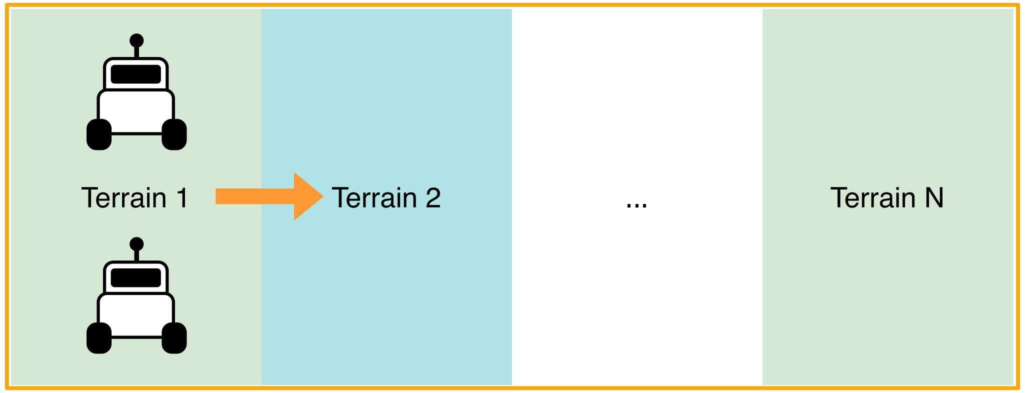
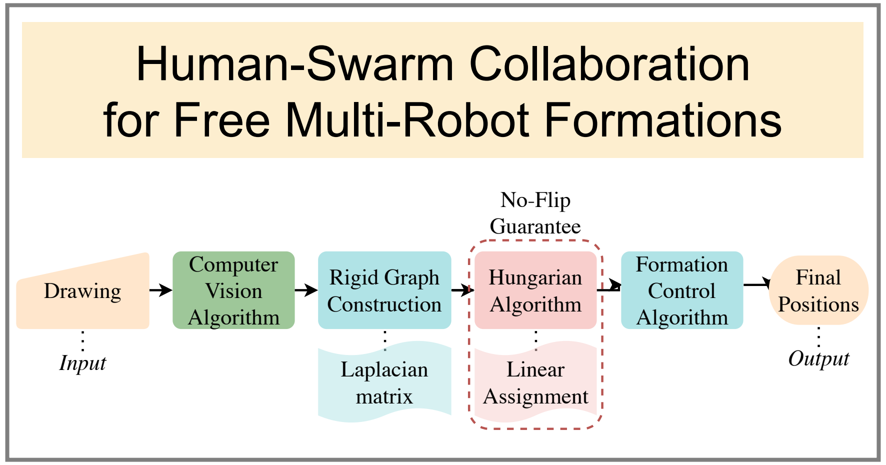
Learning Collaborative Robot Mobility with Model Predictive Scheduling

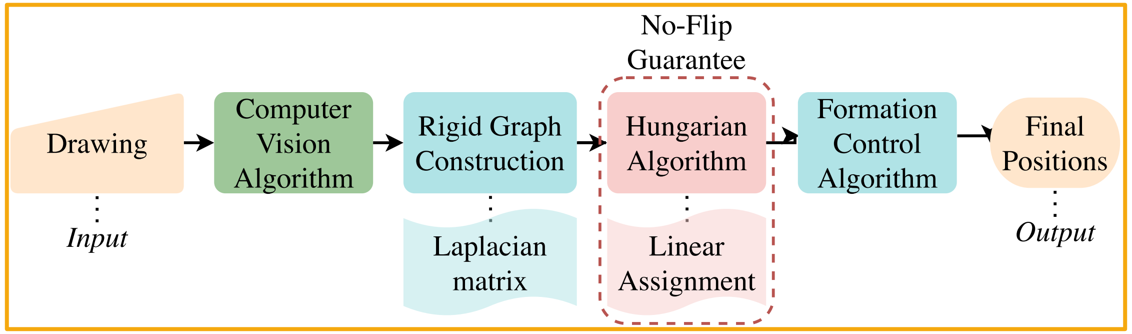
Learning Collaborative Robot Mobility



- Research Idea: Energy-based collaborative scheduling using machine learning
- Motivation: Formulated mixed-integer linear program becomes intractable at scale
- Expected Contribution: Online approximation of the long-horizon collaborative scheduling for energy-efficient mobility


Human-Swarm Collaboration
for Free Multi-Robot Formations
No-Flip
Guarantee
Drawing
Input
Computer
Vision
Algorithm
Rigid Graph
Construction
Laplacian
matrix
Hungarian
Algorithm
Linear
Assignment
Formation
Control
Algorithm
Final
Positions
Output

Human-Swarm Collaboration



- Research Idea: User-friendly graphical user interface that allows non-expert users to specify and adapt swarm formations through freehand sketches
- Motivation: Accessibility, usability, and adaptability of human-swarm collaboration
- Expected Contribution: User study quantifying usability of the interface

Karam, Riwa, "A Graphical Interface for Specifying and Establishing Multi-Robot Formations," MS thesis, University of California, Irvine, 2025.

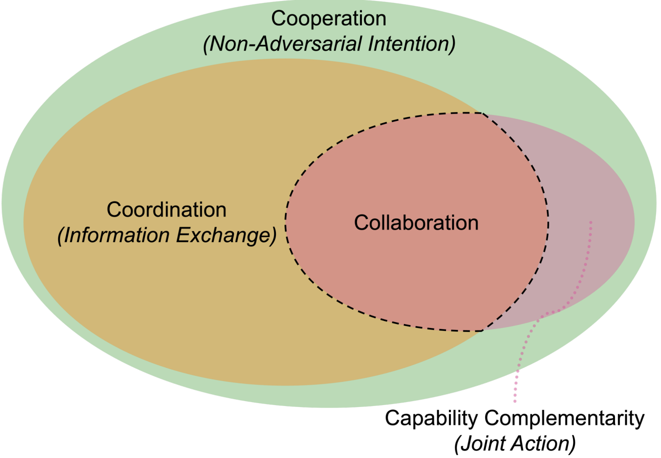
Acknowledgments

Robot Ecology Lab
UCI Samueli School of Engineering

Office of Naval Research
USN
ONR
Science & Technology

UCI Samueli School of Engineering

Collaboration in Multi-Agent Systems [1]



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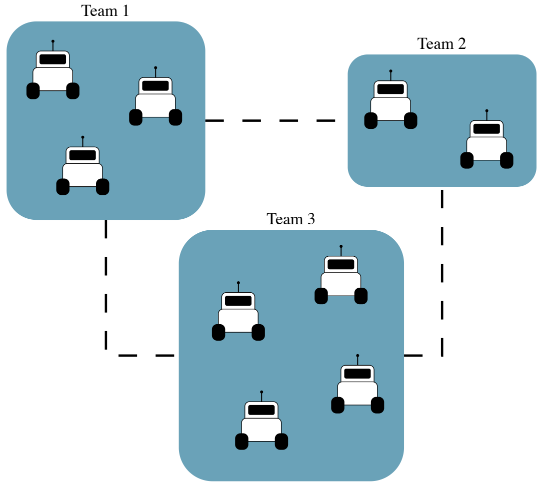

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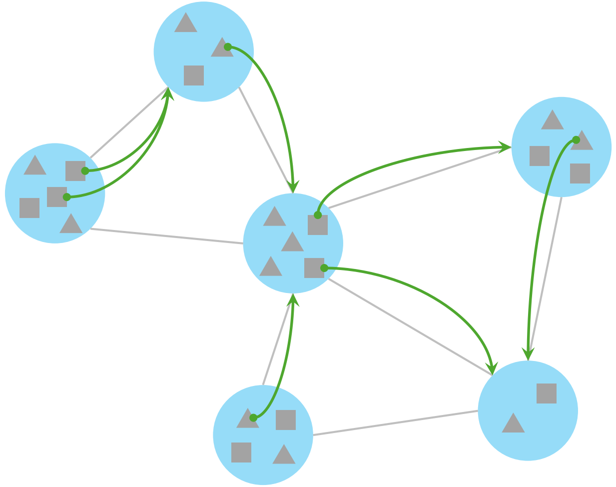
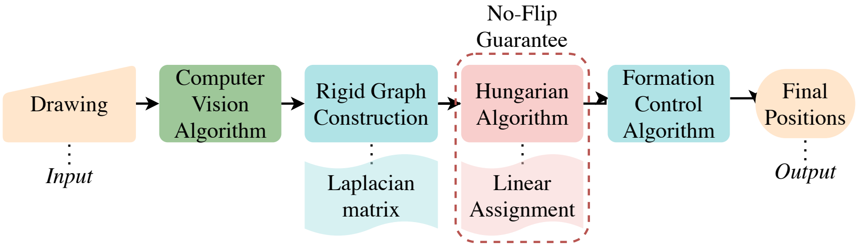


Learning Collaborative Robot Mobility with Model Predictive Scheduling

Thank you!

Human-Swarm Collaboration for Free Multi-Robot Formations



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